

CORAL: Framework for Structuring Individual Cognitive Processes as AI Reasoning Processes

Although mainstream AI development emphasizes efficiency by minimizing human effort, often refining “smart” systems that respond effectively even to vague inputs, this trend paradoxically reduces users to passive operators. Furthermore, at the level of custom instructions, prompt engineering techniques have become increasingly sophisticated; however, their primary objective is to control LLM probabilistic behaviors to efficiently obtain high-quality outputs—a goal distinct from bridging users’ own thinking processes with AI reasoning processes. This asymmetry in technological advancement, weighted toward AI efficiency, implies that the current human–AI collaboration faces a structural constraint in which the thinking processes of users are not adequately reflected in AI reasoning processes. This study proposes an approach to structure and verbalize individual thinking processes, implementing them as AI reasoning processes alongside the CORAL framework that realizes this approach. As a proof of concept, we implemented the S-OPE configuration using CORAL and conducted a comparative evaluation study across the four conditions. In complex problem-solving tasks simulating real-world scenarios (35-point rubric), the S-OPE configuration achieved significantly higher scores than a custom instruction that combines multiple existing prompt engineering techniques (median 30.7 vs. 25.0, effect size $r = 0.83$, $p < 0.001$). Additionally, the ablation analysis confirmed an integration effect that exceeded the sum of the effects of the individual CORAL framework components. This study demonstrates the feasibility of structuring and implementing individual thinking processes in natural languages.

CCS CONCEPTS • Cognitive science • Interaction design theory, concepts and paradigms • User studies

Additional Keywords and Phrases: Human–AI collaboration, End-user development, Metacognition, Cell-based organized reasoning with adaptive linkage (CORAL), transition with inherited discourse exchange (TIDE)

1 INTRODUCTION

1.1 Current limitations in human–AI collaboration

Effective problem-solving inherently draws upon domain expertise, experiential knowledge, and individual cognitive styles—capacities that distinguish expert practitioners from novices. The emergence of large language models (LLMs) has fundamentally transformed human–AI interaction, offering unprecedented computational support for complex tasks. However, this transformation carries a paradox. Mainstream development—the driving force behind current AI advancement—prioritizes efficiency above all, refining “smart” tools that yield high satisfaction even from ambiguous inputs. This efficiency-first paradigm optimizes probabilistic behaviors for general utility, bypassing the user’s cognitive engagement.

The resulting problem is structural: users’ domain expertise and thinking processes are not adequately reflected in AI reasoning processes. This constraint arises from the fact that the existing AI utilization approaches are external tools oriented toward general-purpose functionality. Prompt engineering and existing agent systems have achieved significant results in efficient task support. However, no mechanism exists to integrate individual users’ cognitive

characteristics or problem-solving styles into AI systems. Consequently, users remain in the position of “operators,” and collaboration with AI is limited to standardized interactions.

This challenge is significant from the perspective of human-centered design principles in the field of HCI. Flavell [1] defined metacognition as “thinking about thinking,” emphasizing the importance of being able to monitor and control cognitive processes. Zimmerman’s [2] self-regulated learning theory systematized the process of acquiring expertise through self-reflection. These theories provide a foundation for reflecting the individual cognitive processes in AI systems.

1.2 Approach: structuring cognitive processes as AI reasoning processes

To address the challenges above, this study proposes an approach to achieve effective human–AI collaboration by structuring and verbalizing individual thinking processes and implementing them as reasoning processes for AI agents. This approach enables users to explicitly describe their domain expertise, personal thinking processes, and cognitive styles in natural language and implement them as executable custom instructions for AI.

To realize this approach, we propose a Cell-based organized reasoning with adaptive linkage (CORAL) framework. This framework enables users to implement their thinking processes as AI agent reasoning processes using natural language, allowing users to construct personalized human–AI collaboration systems. Furthermore, as a proof of concept (PoC), we present an implementation example of CORAL and quantitatively demonstrate, through a statistically designed comparative study, that the CORAL implementation exhibits superior performance in complex problem-solving tasks that simulate real-world scenarios.

The “complex problem-solving tasks simulating real-world scenarios” mentioned here do not disclose all information from the outset but reveal information progressively. In real-world problem-solving, all relevant information is rarely available from the start; only through questioning, deepening inquiry, and forming context through dialogue does the full picture of the problem become clear. The evaluation tasks in this study incorporated an information disclosure protocol as a core design element to simulate the complexity of real-world scenarios.

The approach of this study embodies the integration of personalization technologies and end-user development (EUD) in the HCI field, achieving personalization adapted to user cognitive characteristics at a higher level of human thinking and AI reasoning processes. Furthermore, by adhering to Open Science principles and making all the data and custom instructions used in the research publicly available, we ensured transparency and verifiability, paving the way for subsequent researchers.

1.3 Main contributions

This study contributes to the HCI field in the following four aspects:

1. **Proposal for a Structuring and Implementation Approach:** We propose an approach to structure and verbalize individual thinking processes, implementing them as AI reasoning processes within the custom instruction layer.
2. **Construction of the CORAL Framework:** We propose a framework that integrates the division of custom instructions (Cell) with a context inheritance protocol (TIDE) that maintains semantic coherence after division.
3. **Realization of Natural Language End-User Development:** We provide an environment that enables domain experts to implement their thinking processes in AI without technical expertise.

4. **Quantitative Demonstration of Effectiveness:** To evaluate frameworks in the custom instruction layer, we provide a task design example that simulates real-world problem-solving, where information is progressively disclosed through dialogue, and conduct a comparative evaluation study across four conditions ($N = 36$) to quantitatively assess the effectiveness of the CORAL framework.

1.4 Relevance to the HCI community

This study contributes to the HCI readership by demonstrating the feasibility of AI-AI-interactive systems that integrate users’ personal cognitive characteristics and expertise into AI systems, thereby adapting them to user-specific thinking processes.

1.5 Structure of the Paper

The remainder of this paper is organized as follows: Section 2 reviews related work and the theoretical foundations of our approach. Section 3 introduces the CORAL framework and details the S-OPE configuration. Section 4 describes the evaluation methodology used to verify the framework’s effectiveness. Section 5 presents the quantitative results of the comparative study. Section 6 discusses the findings, theoretical implications, and limitations. Finally, Section 7 concludes the paper and outlines future studies.

2 RELATED WORK

2.1 Technical foundations and framework evolution

In maximizing LLM capabilities, reasoning frameworks such as Chain-of-Thought [3], Tree of Thoughts [4], and ReAct [5] have demonstrated the integration of strategic thinking and action. These serve as the technical foundations for CORAL.

Orchestration frameworks such as LangChain [6], AutoGen [7], and CrewAI [8] target efficiency and general-purpose applicability as design goals and assume implementation by programmers. However, the structuring and implementation of individual thinking processes that this study aims to address require different design requirements: domain experts directly implement their thinking processes in natural language.

Technical Differences from Prompt Chaining: The CORAL framework is an extension of prompt chaining but differs technically in the following respects. First, the divisions’ targets differ. While prompt chaining divides “tasks,” CORAL divides “custom instructions (instruction text).” Second, the motivation for division differs. While prompt chaining aims to facilitate efficient task processing, CORAL seeks to prevent attention control breakdowns in LLMs caused by large custom instructions. Third, the inheritance design differs. While prompt chaining passes the previous output as-is or in a summarized form, CORAL inherits a context that is semantically structured across six perspectives. These technical differences give CORAL a logical necessity of “division -> need for inheritance -> TIDE protocol.” The CORAL framework is a form of custom natural-language instruction. However, it is also intended to bring about a qualitative transformation by providing a cognitive scientific structure for custom instructions.

2.1.1 Current Human-AI Collaboration: Structural Limitations and Approaches

Current mainstream human-AI collaboration follows an “external tool paradigm” and is inherently limited in its structure. Tankelevitch et al. [9] demonstrated that generative AI places high metacognitive demands on users across

four stages: prompt formulation, output evaluation, iterative refinement, and workflow integration. This cognitive load stems from the structure of the AI treatment as an external tool. Existing metacognitive support approaches contribute to reducing the cognitive load; however, the framework for viewing AI as “a tool to be operated from outside” remains unchanged.

Similar constraints exist in the integration of EUD and AI. Esposito et al. [10] pointed out difficulties in technical skill acquisition, the black-box nature, and a lack of adaptability. Kelly [11] indicated that establishing evaluation axes to measure true personalization is challenging. The cognitive characteristic-based system adaptation examined by Steichen et al. [12] also remains an “external adaptation” approach.

2.1.2 From External Tools to Structured Collaboration

In response to the challenges identified by these prior studies—the structural limitations of external tool-type collaboration, cognitive load issues, and difficulties with personalization—this study proposes the CORAL framework. In the CORAL framework, users structure and verbalize their thinking processes, implementing them as AI reasoning processes, which enables the design and control of the cognitive load generation structure. While the existing explainable AI aims to improve transparency from the system side, CORAL enables proactive understanding and control from the user side. By adopting a natural language description of thinking processes, it presents a new direction that is distinct from programming-oriented EUD. The next chapter explains the technical details of the CORAL framework.

2.2 CORAL framework

2.2.1 Framework Architecture

The CORAL framework realizes multistage dynamic reasoning processes under a single LLM and session, providing a structure that can flexibly adapt to users’ personal thinking characteristics and task requirements. Specifically, the CORAL framework is a composite of custom instructions called Cells, which have special requirements and stably achieve coordination between Cells through a protocol called TIDE. Cells are explained in Section 2.3, and the TIDE protocol is explained in Section 2.4. Additionally, the CORAL framework incorporates a feedback loop that supports iterative improvements by users. This is explained in Section 2.5.3.

Clarification of Technical Positioning: The CORAL framework fundamentally differs in its design philosophy from multi-agent systems such as LangChain and AutoGen. Although multi-agent systems pursue efficiency and scalability through parallel and distributed processing using multiple LLM instances, CORAL assumes sequential processing within a single LLM and session, with the design goal of maintaining continuity and semantic consistency in task processing. This choice is necessarily derived from the core concept of “structuring individual thinking processes and implementing them as AI reasoning processes.” Deep thinking cannot be parallelized; just as human cognitive processes unfold sequentially, the AI reasoning processes that structure and implement them should also unfold sequentially.

The overall conceptual diagram of the CORAL framework is shown in Figure 1.

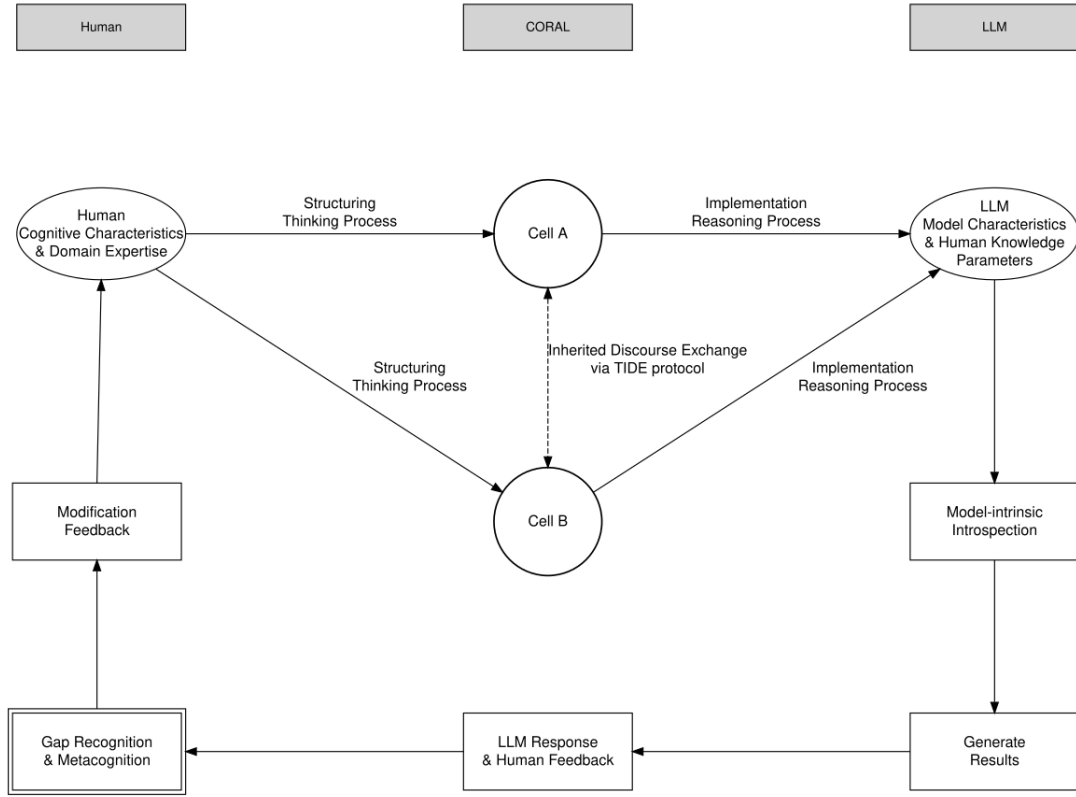


Figure 1: CORAL Framework Conceptual Diagram: This figure illustrates how human cognitive characteristics are structured as Cell-based AI reasoning processes. The left column shows how human domain expertise and cognitive characteristics are structured into Cells (Cell A, Cell B) through the verbalization of thinking processes. The middle column depicts the Cell architecture of the CORAL framework, incorporating the TIDE protocol to enable inherited discourse exchange among Cells. The right column shows how LLM characteristics process inputs and generate outputs. The feedback loop at the bottom enables iterative refinement through gap recognition and metacognitive awareness, supporting continuous improvement of human–AI collaboration.

Cell-based architecture divides complex tasks into manageable functional units, drawing on the principle of the separation of concerns in software development. Each Cell is defined as a custom instruction with a specific role responsible for one cognitive or thinking process. CORAL is not constrained to a specific number of Cells or a fixed configuration. Users can freely add, delete, modify, and connect Cells according to project requirements and the characteristics of their cognitive processes. Various forms of coordination, including layered and circuit structures, are possible.

Examples of flexible Cell-connection patterns in the CORAL framework are shown in Figure 2.

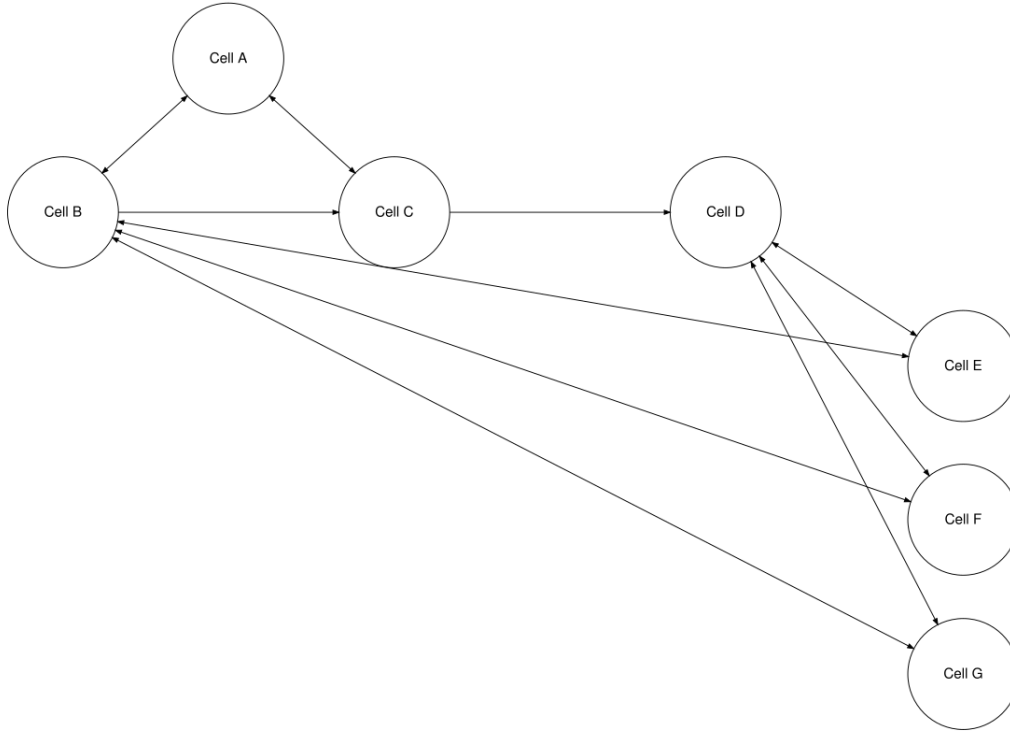


Figure 2: Flexible Cell Connection Patterns in CORAL: Examples of collaborative, hierarchical, and dual-orchestrated configurations. Cells B and D function as Orchestrators, delegating tasks to other Cells based on the nature and complexity of tasks.

The coordination forms between Cells are not constrained to specific architectural patterns and can adapt flexibly to individual thinking processes and tasks with dynamic flexibility. This characteristic is referred to as “Adaptive Linkage” in this study. The adaptive linkage of the CORAL framework is established when the LLM selects its routing based on the TIDE protocol for human-designed reasoning processes and modifies the reasoning process accordingly. Details are provided in Section 2.4. It should be noted that although the CORAL framework is technically capable of diverse implementations, including parallel processing, considering the design philosophy discussed in Section 2.5 and the core concept of “structuring individual thinking processes and implementing them as AI reasoning processes,” sequential routing is assumed for Cell groups handling deep thinking and reasoning.

2.3 Cell: Fundamental building block

2.3.1 Cell Definition and Concept

A Cell in the CORAL framework is a special custom instruction that explicitly assumes the input and output. The differentiating factors from general custom instructions are two design principles: “input-output oriented design” and “explicit reasoning process definition.”

1. Input-Output Oriented Design

- Each Cell was designed with clear input and output assumptions for users and other Cells. This design philosophy enables the division of complex cognitive processes into manageable functional units and standardizes coordination between Cells. Coordination between Cells was standardized using the TIDE protocol described below to achieve consistent context inheritance and state transitions. This standardization enables multiple Cells to function as an integrated system, performing complex tasks while maintaining long-term contextual consistency. A clear information flow was designed, where the inputs received included the inherited context from the previous Cell and user input, and the outputs generated included the inherited context for the next Cell and responses to the user.

2. Explicit Reasoning Process Definition

- Each Cell has its reasoning process described in natural language in the form of “what to do (Do).” This expresses individual thinking patterns and professional judgment criteria as an executable logic for AI. Although conventional custom instructions remain general-purpose directives, Cells implement personalized reasoning flows that reflect the users’ specific cognitive processes and problem-solving approaches.

Cells in the CORAL framework are categorized into two types based on their functions. **General Cells** (Orchestrator, Planner, Executor, etc.) are designed to reduce the users’ cognitive load, whereas the **Sensemaker Cell** is designed to intentionally increase the cognitive load and promote metacognition (thinking about thinking). This dual strategy of cognitive load is detailed in Section 2.5.

The conceptual structure of the Cells in the CORAL framework is shown in Figure 3. As illustrated, each Cell consists of three sections: the Input section receives the inherited context from the previous Cell through the TIDE protocol, the Reasoning Process section executes the Cell’s specific cognitive or processing logic, and the Output section generates an inherited context for the next Cell. The execution flow arrows indicate the sequential processing direction within a Cell.

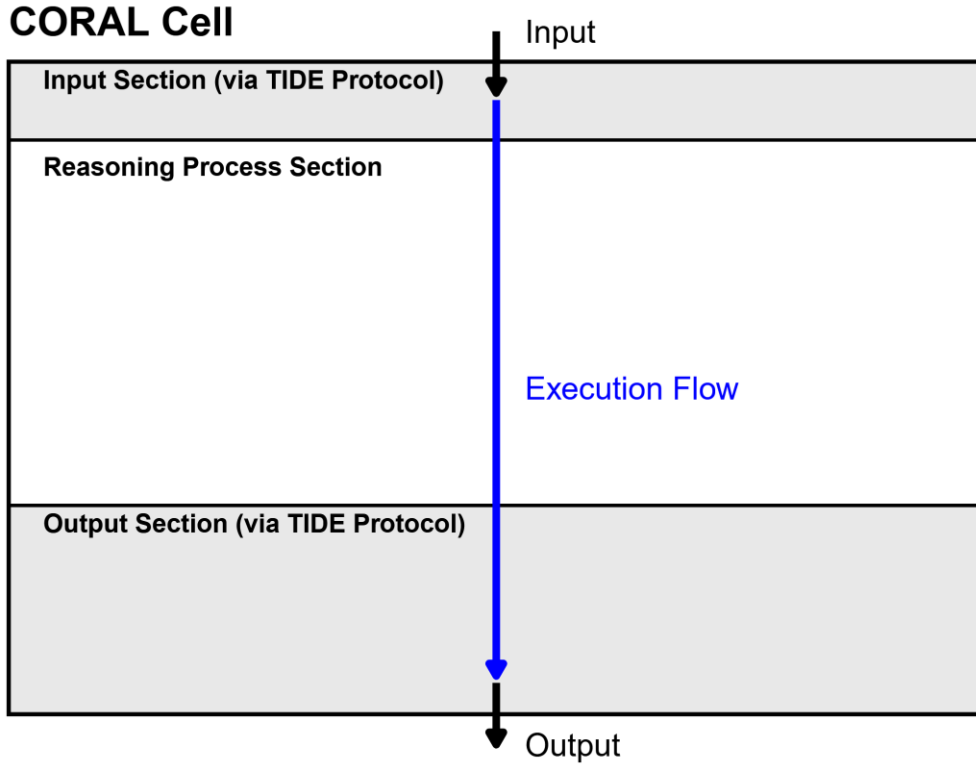


Figure 3: Conceptual Structure of a CORAL Cell: This figure illustrates the input–output-oriented design of a CORAL Cell, highlighting the explicit reasoning process section and the integration of the TIDE protocol that together characterize each Cell in the CORAL framework.

2.3.2 Platform Independence and Technical Implementation

Cells can be implemented on any platform that can explicitly invoke custom instructions. This versatility gives CORAL the potential to develop technologies for structuring and implementing individual thinking processes that are not dependent on specific platforms.

In this study, we adopted the .mdc format in the Cursor environment for Cell definitions and utilized the `fetch_rules` tool for Cell invocation. Each Cell was managed as an independent file, dynamically loaded, and executed as required. This implementation approach facilitates the addition, modification, and deletion of Cells, thereby promoting continuous improvement cycles for users.

For the technical implementation, a Cell file (.mdc) contains the following structured elements: (1) a clear definition of the Cell’s role and responsibilities, (2) the format of information expected as input, (3) a step-by-step description of the reasoning process to be executed, (4) the format of information to be generated as output, and (5) transition

conditions to the next Cell and context to be inherited. This structure enables EUD, where users design and implement AI reasoning processes in their natural language.

The CORAL framework combines multiple Cells to form an integrated system. While maintaining the specialization of individual Cells, the entire complex cognitive process is realized through coordination using the TIDE protocol.

2.3.3 Cell Division: Practical Necessities from Implementation Experience

Cells in the CORAL framework are not a design choice that pursues conceptual elegance or theoretical beauty but rather a necessary choice based on practical constraints that became apparent during implementation. We specifically explain why division into multiple “Cells” as opposed to a single custom instruction was technically unavoidable in realizing the structuring and implementation of individual thinking processes.

In the CORAL framework, the number of descriptions in custom instruction can become enormous when expressing individual thinking processes. For example, in this study, we present the S-OPE configuration (see Section 4) implemented by a developer for paper-writing purposes as an example and quantitative evaluation target. If all the Cells used in the S-OPE configuration were integrated as a single custom instruction, it would result in a massive instruction text of approximately 38,000 characters (in Japanese), comprising approximately 150 processing steps. The instructional text of this scale causes the following fundamental problems:

1. Attention Control Breakdown: The LLM attention mechanism cannot function effectively, and it cannot properly determine which part of the lengthy instruction to focus on. Consequently, instruction compliance becomes unstable, creating difficulty in executing expected reasoning processes.
2. Lack of Maintainability: When modifying parts of an instruction text exceeding 38,000 characters, it becomes extremely difficult to grasp the scope of impact that the change has on other parts of the text. This causes the continuous system improvements to cease functioning effectively.
3. Loss of Reusability: Extracting specific functions (e.g., document creation, code generation, and strategic dialogue) for reuse in other contexts or utilizing them in different combinations becomes practically infeasible. This significantly limits the adaptability to the diversity and evolution of individual thinking processes.

Therefore, Cell division is a necessary design choice based on human cognitive limitations (difficulty in design and maintenance) and the technical constraints of LLM systems (limitations in attention control). Furthermore, to ensure that the divided Cells operate cooperatively while maintaining semantic coherence, it is necessary to develop a TIDE protocol for managing information inheritance and state transitions between Cells and incorporate them into Cells, as detailed in Section 3.3.

2.4 TIDE protocol for context inheritance

The TIDE (transition with inherited discourse exchange) protocol is a mechanism that achieves stable rule switching and context inheritance based on natural language, described in six steps.

Details of the six steps of the TIDE protocol are listed in Table 1.

Table 1: TIDE Protocol Six-Step Framework for Cell Transition and Context Inheritance: This protocol defines the standardized procedure for transitioning between Cells while preserving semantic coherence. Steps 1–3 govern the exit process of the current Cell, including trigger recognition, identification of the next Cell, and specification of critical context. Steps 4–6 manage the state transition declaration and activation of the next Cell, ensuring inherited discourse exchange is properly established.

Step	Name	Description
Step 1	Transition Trigger	Current Cell determines the necessity for transition by recognizing completion of its responsibilities, specific situational changes, or the need for task delegation to another Cell.
Step 2	Next Cell Identification	Identify and select the specific next Cell to transition to based on task nature, complexity, and current context requirements.
Step 3	Inherited Context Specification	Current Cell identifies and organizes critical context information essential for the next Cell’s processing initiation (finalized task objectives, key decisions made, summary of major analysis results, etc.), enabling Inherited Discourse Exchange.
Step 4	State Transition Declaration: End	Current Cell formally declares the end of its role by specifying the ending Cell name, destination Cell name, and critical context information.
Step 5	Next Cell Rule Application	LLM loads the next Cell’s rule file (e.g., *.mdc files in the Cursor environment) using appropriate tools (e.g., fetch_rules in the Cursor environment) and applies it as the new instruction set, immediately beginning processing according to the new ruleset.
Step 6	State Transition Declaration: Start	The newly activated next Cell declares the start of its role by specifying the starting Cell name and begins processing based on the inherited context.

In Step 3 of the TIDE protocol, “Inherited Context Specification” is performed. This is because when achieving state transitions while maintaining and inheriting the task processing context between different Cells, the goal is not merely data transfer but rather transmitting semantic connections, such as the purpose of why analysis was conducted and issues identified from the results, while explicitly emphasizing important information. When the subsequent Cell reads the specified context, an inherited discourse exchange, which is the purpose of the TIDE protocol, is established.

Inherited discourse exchange was conducted from six perspectives: direction, context, significance and priority, relationships and dependencies, abstraction level, and execution and evaluation.

The details of the six perspectives on the discourse exchange are listed in Table 2.

Table 2: Six Components for Inherited Discourse Exchange: This table presents six perspectives that together constitute a comprehensive checklist for context inheritance between Cells. Each perspective is grounded in cognitive science theories: Direction draws on reward-guided learning theory; Context relates to working memory models; Significance and Priority reflect attention and effort theories; Relationships and Dependencies align with mental model theory; Abstraction Level connects to metacognition theory; and Execution and Evaluation are based on cognitive load theory and self-regulated learning theory.

Perspective	Definition
Direction	Purpose of the task (Why) and expected outcomes (What)
Context	Background information and constraints
Significance & Priority	Priority identification and resource allocation
Relationships & Dependencies	Logical connections and dependencies
Abstraction Level	Appropriate granularity of thinking
Execution and Evaluation	Action execution and progress monitoring

1. Direction is defined as a perspective that clarifies the ultimate purpose of the task (why), specific expected outcomes (what), and the main intentions or hypotheses leading to them, forming the foundation for action strategies. This perspective was established based on Rushworth et al.’s [13] neuroscientific research on reward-guided learning and decision-making in the frontal cortex, and Newell and Simon’s [14] problem-solving theory, which realizes action focus through clear goals and intentions, strengthening motivation, and improving planning efficiency.
2. Context is defined as a perspective that clarifies the background information surrounding a task, including its history, the current situation, key stakeholders, available resources, and constraints. This corresponds to understanding the “stage setting” or “preconditions” in which the task is placed. This perspective is based on the context control function of the central executive in Baddeley and Hitch’s [15] working memory model, and Miyake and Shah’s [16] research on the relationship between context processing and working memory in complex cognition. This enables an accurate understanding of tasks, setting appropriate judgment criteria, and clarifying implicit assumptions.
3. Significance and Priority are defined as perspectives for identifying the most essential and impactful elements for achieving current goals among numerous pieces of information and task elements, and concentrating cognitive resources. This perspective is based on Kahneman’s [17] capacity limitation theory regarding attention and effort, as well as Norman and Shallice’s [18] theory of the supervisory attentional system, which plays a crucial role in attention control and priority setting. This theory enables the efficient allocation of limited cognitive resources, prevents information overload, and facilitates expedited decision-making.
4. Relationships and Dependencies are defined as perspectives for grasping the logical “connections” and influence relationships between elements such as individual pieces of information, tasks, and ideas. This corresponds to understanding the “structure” and “interactions” of the element groups that constitute the task. This perspective is based on Johnson–Laird’s [19] mental model theory regarding structural representation and reasoning processes and Gentner’s [20] analogy and structural mapping theory regarding relational mapping, which enables overall picture comprehension and structural understanding, improved planning accuracy, root cause identification, and future prediction.
5. The abstraction Level is defined as a perspective for appropriately selecting and adjusting the granularity of thinking and information (degree of concreteness and abstraction) based on the situation and purpose. This perspective is based on Flavell’s [1] metacognition theory and the levels of processing models in cognitive

psychology [21], which realizes flexible thinking mode selection based on the situation, discovery of essential issues, and effective communication.

6. Execution and Evaluation are defined as perspectives for executing specific actions based on formulated plans and strategies, monitoring and evaluating progress and results against goals (set in “Direction”), and modifying actions or plans as required. This perspective is based on Sweller’s [22] cognitive load theory and Zimmerman’s [2] self-regulated learning theory, which enables the reliable execution of plans, continuous monitoring of goal achievement, and the promotion of learning and improvement cycles from experience.

These six perspectives were discovered inductively during the development process, with their theoretical foundations confirmed through consultation with cognitive science literature. The six perspectives are not completely independent (orthogonal) but mutually complementary. For example, if “Direction” is not established, judgment of “Significance and Priority” becomes difficult, and understanding “Relationships and Dependencies” deepens the grasp of the “Context.” In this study, by treating these perspectives as a comprehensive checklist, we prevented gaps in thinking and achieved a precise contextual inheritance. The exact relationships and weighting of each perspective changed dynamically based on the nature of the task and individual cognitive characteristics. Elucidating these is a topic for future research.

2.5 Design philosophy: Natural language End-User Development, metacognition, and iterative refinement

The CORAL framework is based on the design philosophy of “a mechanism that allows anyone to implement AI adapted to their thinking processes in natural language and continuously improve it.” This philosophy integrates three elements: (1) EUD through natural language; (2) promotion of metacognition; and (3) iterative improvement based on feedback.

Functional Classification of Cells: Cells in the CORAL framework are categorized into two types based on their functions. First, the **Sensemaker Cell** is a specialized Cell designed to promote metacognition (thinking about thinking), posing fundamental questions to users, such as “Why?” and “What is the essential problem?” to encourage deep problem understanding rather than superficial problem-solving. Second, **General Cells** (Orchestrator, Planner, Executor, etc.) are implementations of users’ thinking processes that have been structured, verbalized, and implemented as AI reasoning processes. This classification is based on the dual cognitive load strategy in the S-OPE configuration (see Section 3), intentional cognitive load increase by Sensemaker, and cognitive load reduction by general Cells, as the design foundation. However, the metacognitive promotion effect of Sensemaker is a design intent, and its empirical verification is outside the scope of this study.

2.5.1 Natural Language End-User Development

EUD, which uses natural language, serves as a bridge function for converting individual thinking processes into a form executable by AI. In CORAL, users directly describe Cell-reasoning processes in natural language, without relying on programming languages. This enables domain experts to embed their thoughts into a system without encountering technical barriers. While the inherent ambiguity of natural language carries the risk of misinterpretation, CORAL views this ambiguity as an opportunity to clarify intentions through dialogue with AI and convert tacit knowledge into explicit knowledge (see Section 2.5.2).

2.5.2 Metacognitive Enhancement through Description Process

Promoting metacognition is one of CORAL's design intentions. As Tankelevitch et al. [9] analyzed, in conventional AI use, users are forced to engage in advanced metacognitive activities of prompt formulation, output evaluation, and strategy modification. In CORAL, users structure their thinking patterns according to Cell rules. This Cell description act is structurally similar to Flavell's [1] "metacognition"— "thinking about thinking." Users gain opportunities to view their cognitive processes objectively through descriptions. Additionally, by structuring and implementing thinking processes as Cells, it is expected that the load of prompt formulation will be transferred to Cell rules, and output evaluation will change to "consistency confirmation of internalized thinking."

The design Intent was based on Schon's [23] reflective practice theory. CORAL's Cell design, description, and improvement cycle is designed to support the circulation of "knowledge in action" and "reflection on action." Users observe the AI behavior, recognize gaps in their intentions, and modify their Cell descriptions. Unintended AI responses are reframed not as "poor AI performance" but as "ambiguity in the thinking process description." Through this reflective practice loop, users gain opportunities to convert tacit knowledge into explicit knowledge (see [Section 2.5.3](#)). However, these metacognitive promotion effects are theoretical assumptions based on design intent, and empirical verification was not conducted in this study. Longitudinal measurement of these effects is a topic for future research.

2.5.3 Iterative Refinement Through Feedback Loop

The form realized by the CORAL framework is a feedback loop in which users observe AI responses, improve Cells, and observe the results again. Cells described in natural language function as a "common thinking blueprint" that is understandable and editable by both humans and AI. Users observe AI responses and improve the Cells to adjust their AI to a system that is adapted to their cognitive characteristics. Simultaneously, through dialogue with the AI, users gain opportunities to view their thinking processes objectively and convert tacit knowledge into explicit knowledge.

This improvement cycle can be understood from the perspective of Zimmerman's [2] self-regulated learning theory and Argyris and Schon's [24] organizational learning theory (single-loop and double-loop learning). The CORAL framework technically supports users' SRL processes. Specifically, the act of users designing and implementing their thinking processes as Cell rules corresponds to the "forethought phase" in self-regulated learning. The processes where the AI agent executes tasks based on implemented Cell rules, and users observe the results, correspond to the "performance phase" and "self-monitoring phase," respectively.

Users evaluate the AI agent responses and artifacts according to their intentions, expectations, and task goals (self-judgment). Based on this evaluation, if a gap exists between expectations and the AI agent's behavior, users modify and refine Cell rule descriptions or add or delete new Cells to adjust the AI's reasoning process (self-reaction). This series of user "self-monitoring -> self-judgment -> self-reaction" cycles functions as practical "feedback" for the AI, making AI behavior more adapted to users' thinking styles and task requirements.

The depth of "self-reaction" here determines the quality of improvement. Based on Argyris and Schön's theory, when user reactions remain focused on fine-tuning Cell rules or clarifying specific instructions, this corresponds to "single-loop learning." This advances the existing frameworks. However, through repeating this single-loop learning, or through unexpected insights gained from dialogue with AI agents, or through the process of articulating one's thinking processes as Cells (promotion of metacognition), users may arrive at more fundamental questions, such as "Is this division of Cell roles really appropriate?" "Is the current form of information coordination between Cells using the TIDE protocol suitable for resolving essential problems?" At this stage, users may reconsider the design philosophy of the existing Cells, add

Cells with new roles, or significantly change the coordination structure between the Cells and the rules of information inheritance. This corresponds to “double-loop learning,” accompanied by transformation of the thinking framework itself.

Thus, the CORAL framework has a structure that can be developed from single-loop learning to double-loop learning, according to the depth of user reflection based on a self-regulated learning cycle. Each time users improve the Cell rules, the AI agent deploys more refined reasoning according to the updated “thinking blueprint,” and the precision and efficiency of task execution improve. Through this continuous feedback loop and occasional fundamental reviews, users enhance their metacognitive abilities to objectively view and structure their thinking processes. Consequently, a continuous improvement cycle based on feedback was implemented.

2.5.4 Inherent Difficulty of Thought Process Structuring

The challenge in implementing the CORAL framework lies in the fundamental difficulty of externalizing and structuring one’s thinking processes in a natural language. This difficulty stems not from superficial issues such as technical implementation complexity or tool usage; rather, it arises from the essential characteristics of human cognitive processes, particularly those attributable to the unconscious nature of thinking processes. Most everyday thinking and problem-solving processes are executed automatically and unconsciously [25]. For example, processes in which experts intuitively grasp problems and find solutions, the basis for experience-based judgments, and criteria for priority decisions often exist as tacit knowledge that individuals themselves cannot clearly verbalize. To articulate these thinking processes as Cell rules, metacognitive activity is required to become aware of one’s unconscious cognitive processes. Properly executing this “thinking about thinking” requires cognitive load and time investment, which can become a psychological barrier for many users.

This barrier paradoxically indicates the value of the CORAL framework. By directly addressing the essential difficulty of metacognition, the CORAL framework finds its purpose in achieving a deep level of personalization that is difficult to achieve using conventional approaches.

2.6 Attention control mechanisms in large language models

The sections thus far have primarily explained the design philosophy and human-centered aspects of the CORAL framework. This section explains how the CORAL framework functions in terms of the technical aspects of LLM attention mechanisms and achieves stable reasoning. This technical understanding provides the foundation for understanding the specific implementation and evaluation of the “S-OPE configuration,” an implementation example of the CORAL framework, in the subsequent Section 3.

The CORAL framework can be understood as a structure for explicitly and stably executing attention control by utilizing the attention mechanism of the transformer models. In the CORAL framework, complex and advanced custom instruction groups are divided into smaller, manageable functional units called “Cells,” while defining and dividing the tasks handled by each Cell. This enables the LLM to focus only on “the task it should handle now” in each Cell, making it easier to maintain attention [26].

In the TIDE protocol, rules are established for transmitting particularly important information as “inherited context” when transitioning from one Cell to the next, allowing the LLM to accurately understand “what to focus on next” based on the processing results of the previous Cell, and continue tasks while maintaining attention without losing context. Simultaneously, the TIDE protocol structurally changes the reasoning process of the LLM through dynamic custom

instruction loading. As each Cell defines different thinking patterns (stage structure, judgment criteria, output format, and information processing priorities), the thinking framework is switched.

Each Cell’s reasoning process design is a structure and verbalization of human thinking processes implemented as AI reasoning processes that process information, make judgments, and generate outputs based on patterns unique to that Cell. Through this technical implementation, the CORAL framework causes multiple specialized reasoning processes to function in an integrated manner within a single LLM and session while maintaining semantic coherence.

Based on the explanation of the theoretical foundation and technical characteristics of the CORAL framework presented thus far, the next chapter details the Sensemaker–Orchestrator–Planner–Executor (OPE) architecture, an implementation example demonstrating the implementation and effectiveness of this framework. This PoC concretely demonstrates how the core concept of the CORAL framework, structuring and implementing individual thinking processes, functions, and demonstrates its effectiveness in actual research and development situations.

3 IMPLEMENTATION: S-OPE

3.1 S-OPE implementation: Overview

In this study, we designed and implemented the S-OPE (Sensemaker-Orchestrator-Planner-Executor) configuration as an example and a quantitative evaluation target of the CORAL framework.

The S-OPE configuration consisted of four Cell types. The role of each Cell type is listed in Table 3. Sensemaker promotes essential problem inquiry through strategic dialogue; the Orchestrator serves as the central coordination hub for task classification and routing; the Planner develops systematic execution strategies; and the Executor (comprising Task Manager, Writer, Coder, and Runner) handles specialized domain execution.

Table 3: S-OPE Implementation Components: This table presents the four Cell types along with their specific implementations and functional roles within the S-OPE configuration for structuring and implementing thinking processes.

Cell Type	Implementation	Functional Role
Sensemaker	00_sensemaker.mdc	Strategic dialogue and essential problem exploration through productive cognitive difficulties, promoting deep reflection and insight generation for complex challenges requiring fundamental reconceptualization
Orchestrator	01_orchestrator.mdc	Central coordination hub implementing separation of concerns principles: instruction interpretation, task classification, initial context gathering, and intelligent routing to appropriate execution paths
Planner	02_planner.mdc	Hierarchical task network planning and systematic execution strategy development based on inherited context from Orchestrator or Sensemaker, employing structured decomposition and dependency analysis
Executor	03_a_task_manager.mdc	Specialized domain execution across four implementations Task Manager (cross-executor orchestration & progress monitoring),
	03_b_writer.mdc	Writer (document creation and editing),
	03_c_coder.mdc	Coder (software development),
	03_d_runner.mdc	Runner (terminal command execution)

A conceptual diagram of the S-OPE configuration is shown in Figure 4.

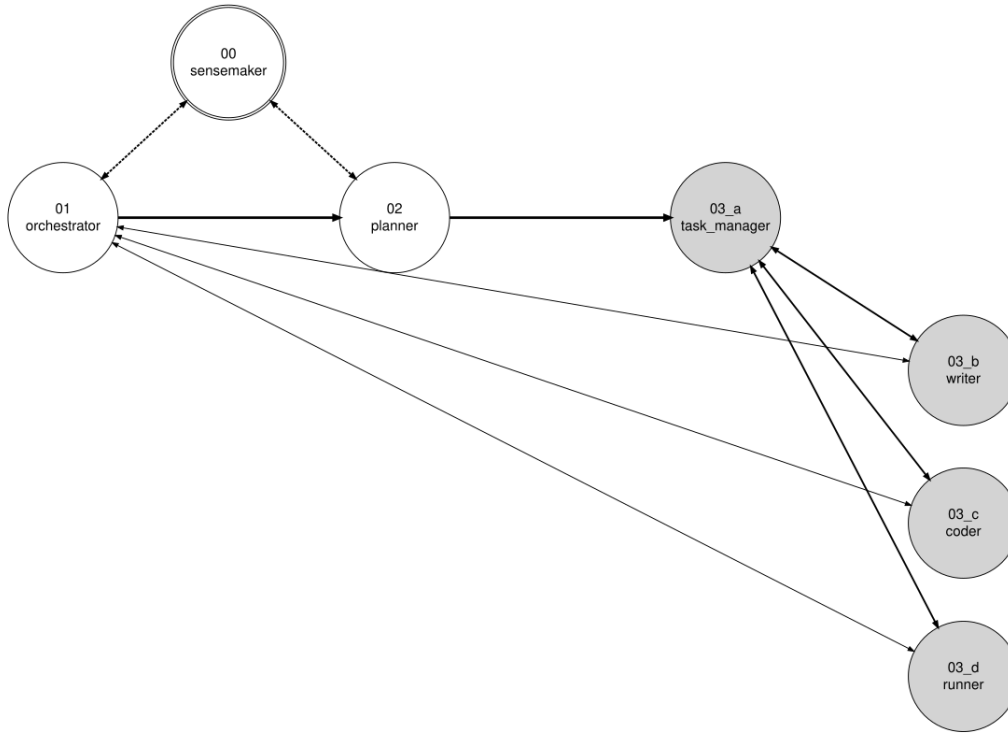


Figure 4: S-OPE Architecture: This figure illustrates the four-Cell configuration implementing a dual cognitive load strategy. The Sensemaker (00, double-bordered) provides strategic dialogue that promotes metacognition through productive difficulties. The primary workflow (Orchestrator->Planner->Task Manager) reduces cognitive load through systematic task decomposition. The Task Manager (03_a) orchestrates specialized Executors: Writer (03_b), Coder (03_c), and Runner (03_d), shown in gray. Thick lines indicate the primary workflow, whereas dashed lines represent optional strategic consultation pathways.

This configuration represents the structuring and verbalization of the thinking pattern “essential inquiry -> overall comprehension -> plan formulation -> execution” that the developer developed through pharmaceutical research and practical experience in private companies, implemented as AI reasoning processes.

The S-OPE configuration adopts a design philosophy based on cognitive load theory [22]. According to Sweller’s cognitive load theory, human working memory has limitations, and the appropriate management of cognitive load determines the efficiency of learning and problem-solving.

However, in S-OPE, rather than a “cognitive load reduction” approach, a dual strategy is adopted as a design intent: reducing unnecessary cognitive load while intentionally maximizing cognitive load related to essential thinking. Specifically, each Cell—Orchestrator, Planner, and Executor—is designed to efficiently reduce the cognitive load during task classification, organization, and execution, thus eliminating the need for users to allocate attentional resources to complex procedures and implementation details. However, the Sensemaker Cell is designed to intentionally increase cognitive load, requiring users to think fundamentally and strategically. This involves asking questions such as “Why?,” “What is the essential problem?,” and “Are the assumptions appropriate?” to promote deep problem understanding rather than superficial problem-solving.

This allocation of cognitive load is based on the concept of ‘desirable difficulties’ [27] in cognitive science. The strategic dialogue process provided by the Sensemaker Cell is intended as a technical implementation of these desirable difficulties. Note that, although the effectiveness of the Sensemaker Cell was statistically confirmed through ablation analysis (see Section 6.1), the mechanism of its effect (metacognition promotion, introduction of desirable difficulties, etc.) was not verified in this study.

3.2 S-OPE implementation: Use cases

The use cases of the S-OPE configuration, extracted from the actual usage experience during CORAL development and paper writing, are listed in Table 4. These use cases are practical examples in which Cell transition patterns dynamically change according to task complexity and nature, and cognitive load allocation strategies were flexibly applied. For simple tasks, direct execution minimizes the procedural cognitive load (Use Case 3); for complex tasks, stepwise planning and execution achieve gradual cognitive load reduction (Use Case 2); and for strategic challenges, productive difficulties are intentionally introduced through dialogue with the Sensemaker (Cases 1 and 4). The S-OPE configuration implements the CORAL framework’s *adaptive linkage* concept, tailored to support developers’ cognitive workflows during demanding tasks such as framework development and academic writing.

Table 4: S-OPE Use Cases: Cell Transition Patterns Across Different Task Complexities. These patterns demonstrate how the CORAL framework dynamically adapts routing strategies based on task complexity. Simple tasks follow direct routes (01->03_b) for efficiency; complex tasks involve planning stages (01->02->03_a->...); strategic challenges engage the Sensemaker for essential inquiry (01->00->01->...); and error scenarios trigger hierarchical escalation to support robust recovery.

Use Case	Cell Transition Pattern	Key Value
Concept Development	01->00->01->02->03_a->03_b->03_a->03_d	Essential problem exploration through productive difficulties, enabling deep cognitive partnership
Complex Technical Implementation	01->02->03_a->03_c->03_a->03_c->03_a->03_b	Systematic approach to complex tasks with iterative refinement and progress monitoring
Simple Document Modification	01->03_b	Cognitive load reduction through intelligent task classification and direct routing
Error Escalation & Strategic Re-planning	01->02->03_a->03_c(error)->03_a(escalation)->01->00->01	Robust error handling through hierarchical escalation and strategic re-thinking, demonstrating system resilience

These use cases are examples that occurred during the process of writing this paper using the S-OPE configuration.

In concept development, productive difficulties are intentionally introduced through deep dialogue with Sensemaker, promoting essential problem inquiries rather than superficial solutions. The Cell transition pattern, including the Sensemaker (01->00->01->02->03_a->03_b->03_a->03_d), demonstrates adaptive routing, where, when strategic thinking is required, the Orchestrator recognizes the nature of the task, selects consultation with the Sensemaker, gains deep insights, and then transitions to planned execution.

For complex tasks (document writing and technical implementation), a systematic cognitive load reduction was achieved through stepwise planning and iterative improvements. This Cell transition pattern (01->02->03_a->03_c->03_a->03_c->03_a->03_b) is commonly used in S-OPE configurations, where the Planner and task manager collaborate to divide the CTs into manageable units, and iterative coordination with the Coder improves the quality. This pattern

demonstrates the structuring capability of the CORAL framework to process technical complexities in a form adapted to human cognitive capabilities.

For simple document modifications, direct execution (01->03_b) minimizes the procedural cognitive load, freeing users from nonessential work. When the Orchestrator recognizes task simplicity, it bypasses the planning stage and delegates it directly to the writer, achieving both efficiency and promoting cognitive resources.

Error escalation with strategic re-planning is the most complex pattern, demonstrating system robustness, where errors at the execution level are guided to strategic reconsideration through appropriate escalation to higher layers. This demonstrates that the CORAL framework can adapt to unexpected difficulties.

3.3 S-OPE implementation: Development process

The S-OPE configuration implemented and evaluated in this study was developed to prove the concept of the CORAL framework, and the current configuration was established in parallel with framework formation. The S-OPE configuration development process shown below is based on a post hoc analysis of records that naturally accumulated during development, not on systematically planned data collection. The development process for the S-OPE configuration is summarized in Table 5.

Table 5: S-OPE Development Process: Scale Progression (1->7 rules) and Architectural Evolution Over Five Months. The development shows iterative refinement from a single coding rule to a complete seven-rule framework. Key milestones include the introduction of strategic dialogue (April 2025), the addition of orchestration (April 2025), an integration experiment failure that prompted the completion of Cell structure and emergence of the TIDE protocol (May 2025), and the finalization of the complete TIDE protocol with six-perspective inherited context definitions (June 2025).

Development				
Phase	Configuration Scale	Key Technical Innovation		Architectural Evolution
Initial (Jan 2025~)	1 rule, 3.4K characters	Basic coding		Single-purpose tool
2025-04-16	2 rules, 6.5K characters	Strategic	dialogue	Dual-mode
(Dialogue)		Multi-instruction		coordination
2025-04-27	4 rules, 12K characters	Dynamic	switching	Multi-agent
(Orchestration)		Cell-like architecture		orchestration
2025-04-28	3 rules, 15K characters	Integration	attempt	Failed integration ->
(Integration)		(stability reduced)		learning
2025-05-09	7 rules, 36K characters	TIDE protocol	emergence	Modular recovery
(Cell Structure)		Cell structure	complete	Near-current form
		(inheritance undefined)		
2025-06-05 (TIDE Complete)	7 rules, 38K characters	6-perspective	context	Stable final system
		TIDE complete		

Based on the recorded development processes, implementation began in January 2025, and quantitative expansion from one rule (3.4K characters) to seven rules (38K characters) was observed over approximately five months. Major technical changes at each stage include the introduction of strategic dialogue functionality (April 16, 2025), the addition of orchestration functionality (April 27, 2025), integration experiment and its failure (April 28, 2025), Cell structure completion with TIDE protocol emergence (May 9, 2025), and complete TIDE framework implementation (June 5, 2025).

In particular, from April 28, 2025, to May 9, 2025, following the failure of the integration experiment, the TIDE protocol was first introduced, and the Cell structure was subsequently completed. During this period, the description volume increased approximately 2.5 times from 15K characters to 36K characters. However, the six perspectives of inherited context have not yet been defined.

The final definition of the six perspectives was completed on June 5, 2025, establishing the complete TIDE protocol and S-OPE configuration, which became the target of quantitative evaluation.

Section 4 describes the statistical experimental design for quantitatively evaluating the performance that the S-OPE configuration actually demonstrates on complex tasks.

4 EVALUATION FRAMEWORK

4.1 Research design and comparative methodology

This study adopted a first-person research approach in which the developer served as the evaluator. In the field, when designing and evaluating novel interactive systems, introspective insights gained from the designer’s deep engagement with the system are indispensable (Schn 2017). In particular, given the nature of the CORAL framework, structuring and implementing individual thinking processes, a PoC targeting the developer’s cognitive processes is the most direct method for demonstrating framework feasibility.

Accordingly, this study prioritizes internal validity over external validity at the PoC stage. Employing an intermediate sample size (e.g., 3–5 users) would introduce individual differences in metacognitive readiness and domain expertise without providing sufficient statistical power to adequately model these confounding factors—resulting in the disadvantages of both small- and large-sample designs. A properly powered multi-user study, by contrast, constitutes confirmatory rather than exploratory research. The single-developer design therefore isolates the architectural contribution of the framework by holding user-level variables constant. Although this methodological choice necessarily limits external validity (generalizability) (see Section 6.4), it is appropriate for a PoC study in underexplored research domains and establishes a foundation for subsequent validation with more diverse user populations.

To verify the effectiveness of the CORAL framework, we conducted a controlled quantitative evaluation within the framework of statistical hypothesis testing.

All tasks in this evaluation were conducted using Cursor (version 1.1.3 to 1.2.1), using a single LLM (claude-4-sonnet thinking model, released 2025-05-22). The features that could affect the behavior of the AI agent in the IDE cursor were restricted. Specifically, the rules for AI (special custom instructions effective for all projects) contained only instructions regarding response language (always responding in Japanese). Memories (a function to save and rule-ify useful context from chat) and To-Do List (a function to generate task lists for AI agents to track tasks) were prohibited.

A complex, high-difficulty task was set as the challenge, and reports (final artifacts) were scored through a rubric evaluation by a dedicated scoring AI agent, which compared score distributions. The overall design of the quantitative evaluation is illustrated in Figure 5.



Figure 5: Experimental design for evaluating the CORAL framework. Four conditions used in the evaluation were compared. The test condition employs the complete S-OPE configuration. The control condition uses integrated custom instructions combining Chain-of-Thought (CoT), Reasoning and Acting (ReAct), and Self-reflection techniques. The Ablation-S condition removes the Sensemaker Cell to evaluate the contribution of strategic dialogue. The Ablation-H condition removes the TIDE-based context inheritance definitions to evaluate the contribution of context inheritance. Each condition includes $N=9$ trials, evaluated by a scoring AI agent using a seven-item rubric (35 points maximum). Statistical power was set at 81.0% with $\alpha=0.05$ and false discovery rate (FDR) adjustment.

4.2 Evaluation tasks and complexity design

In this study, the “report” was used as the final artifact from high-difficulty tasks and scored using a pre-established rubric.

4.2.1 Task: Business Challenge

Designed as a problem discovery and plan modification task in a new product launch plan, it evaluates the following capabilities: (a) ability to formulate structured plans from ambiguous initial instructions; (b) ability to respond to severe constraint conditions added midway (budget cuts) and make realistic plan modifications; and (c) ability to autonomously

detect contradictions in the presented information (conflict between mandatory functional requirements and legal regulations), raise issues, and propose logical examinations and solutions.

In the context of the scenario, extensive deep information, obtainable through questioning, was established and designed to enable the achievement of a 5-point evaluation through appropriate information extraction. Following the information disclosure protocol, information was progressively disclosed based on the AI response level and dialogue depth.

4.2.2 Information Disclosure Protocol

The above tasks were intentionally set as high-difficulty tasks that directly test advanced cognitive processes such as “progressive complex information inheritance,” “long-term contextual consistency maintenance,” and “problem redefinition and refinement through dialogue.” This study aims to evaluate the characteristics of the CORAL framework from multiple angles and faithfully simulate complex real-world problem-solving processes. To improve the reproducibility and fair evaluation of AI capabilities, the following hierarchical information disclosure protocol was adopted for the tasks in this study:

Level 1 (Basic Information): Basic facts, background information, and term definitions necessary for task execution. Disclosed when AI makes direct, specific questions about particular facts or basic situations.

Level 2 (Analytical Information/Background Issues): Underlying complex issues, potential conflicts of interest between stakeholders, and detailed situations that may influence decision-making. Disclosed when AI goes beyond information exploration to connect multiple pieces of information for situational analysis, presents questions or hypotheses based on deeper insights, or asks about hidden assumptions or causal relationships beyond the information presented.

Level 3 (core information/deep psychology/strategic information): Important undisclosed information related to task essence, deep psychology, and true intentions of characters, confidential information affecting strategic judgment, and important constraint conditions intentionally concealed. When AI attempts to identify the root causes of problems, it integrates multiple complex pieces of information to suggest essential problem-solving directions, ask questions that approach the core of ethical dilemmas, or infer the true motives and intentions behind the characters’ actions and statements, demonstrating extremely deep insight, analysis, and questioning.

4.3 Evaluation rubrics and scoring protocol

4.3.1 Seven-Item Evaluation Rubric

Reports were evaluated on the following seven items, scored on a 35-point maximum scale with 0-5 points for each item.

1. Essential Problem Identification and Structuring Capability: Ability to capture essential problem structures from multiple angles in complex information and situations, clearly define problem cores, and effectively redefine the problems to be solved when necessary.
2. Adaptive Analysis Capability: Ability to flexibly adjust and deepen analysis and proposals in response to changing constraints (budget, new information, etc.).
3. Multiperspective Consideration Capability: Ability to gain insights from diverse perspectives (technical, ethical, social impact, etc.) and consider potential risks and biases.

4. Integrated Impact Evaluation Capability: Ability to evaluate the multifaceted impacts of proposals and choices (ethical, planning, stakeholder impacts, etc.), consider trade-offs, and support rational decision-making.
5. Practical Solution Proposal Capability: Ability to deeply understand the task background and context (stakeholders, constraints, objectives, etc.) and formulate and propose specific, executable solutions.
6. Value-aligned Output Capability: Ability to produce final proposals and artifacts that are aligned with basic project/organization purpose values (mission, ethics, etc.), and are logical and of high quality.
7. Logic and Information Reliability: Generated reports are described plainly and accurately, based on a clear logical structure, with the reliability and validity of the referenced information sources guaranteed.

Each item was scored on a 0-5 point scale according to detailed meta-criteria (0: insufficient, 1: Needs Improvement, 2: partial, 3: Good/As Expected, 4: excellent, and 5: outstanding). Complete rubric definitions, including scoring criteria for each level, are available in the Zenodo Repository [28].

4.3.2 Scoring AI Agent Operating Protocol

To minimize evaluation bias, scoring was conducted using an artificial intelligence scoring agent designed in advance for each task, following the protocol:

- Evaluation Session Independence: To eliminate bias from the evaluation history, the evaluation starts in a new chat session.
- Information Restriction: Only the “Report” to be evaluated and related task definitions are provided as information.
- Condition Blinding: Condition identifiers use symbols (t, c, as, ah) to prevent the scoring of the AI agent from inferring the condition meaning from the names. Combined with the information restriction mentioned above, the scoring AI agent evaluates reports without knowledge of the experimental conditions that produced them, achieving a single-blind evaluation with respect to the AI evaluator.
- Three evaluations with average-value adoption: Three evaluations were conducted for each evaluation target, and the average value was used as the final artifact score. The average values were processed to the second decimal place according to IEEE 754, and the first decimal place value was adopted.
- Transparency Assurance: Chat histories of all evaluation processes (evaluation chat logs) were recorded and unedited.

4.3.3 Comparison/Control Conditions

The following four conditions were used for the systematic comparative evaluation. In designing the ablation conditions, we targeted two core functions of the CORAL framework: the strategic dialogue function of the Sensemaker Cell and the context inheritance function of the TIDE protocol. The reason for selecting these two rather than other Cells, such as Orchestrator or Planner, is that Sensemaker is a representative example of “functions impossible to realize with a single prompt” (see Section 2.3.3), and the TIDE protocol is an essential mechanism for maintaining semantic coherence after Cell division. These two components directly support the purpose of the CORAL framework, and we determined that identifying their contributions was crucial for verifying the effectiveness of the framework.

1. Test Condition: As S-OPE Configuration uses a complete implementation of the CORAL framework composed of multiple advanced custom instruction groups (Sensemaker, Orchestrator, Planner, Executor).

2. **Control Condition:** As Integrated Custom Instruction uses a single advanced custom instruction integrating a Chain of Thought (CoT), Reasoning and Acting (ReAct), and self-reflection elements. This condition functions as a representative of existing advanced prompt engineering techniques, providing a strong baseline. Specifically, it is a custom instruction of approximately 2,000 characters, including a 5-stage structured thinking process, seven self-evaluation items, explicit requirements for critical thinking (bias detection and logical leap elimination), and a 3-part fixed output format—a design with up to 4–5 times the technical density of typical official guidelines (approximately 200-500 characters). Note that the Control condition does not include strategic dialogue functionality such as that provided by the Sensemaker Cell. This is not an arbitrary omission but is due to technical constraints. If the Sensemaker Cell (approximately 4,500 characters) was integrated into the Control condition (approximately 2,000 characters), it would become a single custom instruction of approximately 6,500 characters or more, causing an attention control breakdown (see Section 2.3.3). This technical constraint suggests the purpose of the CORAL framework, which enables functions that are impossible with a single prompt, through Cell division.
3. **Ablation-S Condition:** As S-OPE Configuration (Sensemaker Cell Missing) uses the S-OPE configuration with the Sensemaker Cell removed. This identifies the contribution of strategic dialogue to essential inquiries.
4. **Ablation-H Condition:** As S-OPE Configuration (Handover Context Definition Missing), uses the S-OPE configuration with all custom instruction descriptions related to the `{{handover_context}}` definition and use between the Planner Cell and Executor Cell removed. This identifies the contribution of context inheritance functionality using the TIDE protocol.

4.3.4 Counterbalancing Using Latin Square Design

To minimize order effects, such as learning and fatigue effects arising from the experimenter’s prompt input during task execution, the implementation order of the four conditions (Test, Control, Ablation-S, and Ablation-H) was counterbalanced using a Latin square design. Specifically, each condition was assigned different implementation positions (1st through 4th) as equally as possible across the entire trial set of $N = 36$ (nine trials $\times$ four conditions).

4.4 Statistical analysis and power analysis

4.4.1 Primary Endpoint

This study used the report score distribution as the primary evaluation indicator to statistically test whether there were significant differences in the distribution of seven-item evaluation scores (0–5 points per item, totaling 35 points) scored by the scoring AI agent based on the rubric.

4.4.2 Secondary Endpoints

1. **Median of Excellent Evaluations (4-point) Achieved in final artifact scores:** The median of the number of 4-point scores achieved among the seven items of the final artifact scores between conditions.
2. **Median of Outstanding Evaluations (5-point) Achieved in final artifact scores:** The median of the number of 5-point scores achieved among the seven items of the final artifact scores between conditions.

4.4.3 Statistical Methods and Sample Size Design

For statistical comparison, the Mann–Whitney U test (also known as the Wilcoxon rank-sum test) was employed, given the small sample size and ordinal scales. This non-parametric method does not assume a normal distribution, is robust to outliers, and functions appropriately with small sample sizes. It tests for differences in medians and provides practical and easy-to-interpret comparisons.

4.4.4 Power Analysis for Optimal Sample Size Determination

For a controlled statistical comparison, a simulation-based power analysis was implemented in advance. Statistical power (Type II error probability) was pre-calculated to determine the appropriate sample size. The goal was to detect a median improvement of 3 points (assumed median of 28 points) compared to the Control condition's "final artifact score" (assumed median of 25 points) as the minimum practically meaningful effect. This 3-point improvement corresponds to the S-OPE configuration achieving an average of four points (excellent) in a task, where the final artifact score average of the control condition was 3.57 points (four items at four points, three items at three points), representing a non-negligible difference in actual problem-solving.

The following assumptions were set in the power analysis:

- Primary Analysis: Equal variance ($\sigma=2.0$) was assumed for both groups—Control group (mean 25 points, standard deviation 2.0 points) and S-OPE group (mean 28 points, standard deviation 2.0 points)—using a model that rounds values generated from a normal distribution to integers and clips within range.
- Sensitivity analysis: A truncated normal distribution is used to impose more realistic constraints. Control group (mean 25 points, standard deviation 2.0 points, truncation range {18, 30}), S-OPE group (mean 28 points, standard deviation 2.0 points, truncation range {21, 33}).

Through 10,000 simulations, it was confirmed that $N = 9$ per group achieved 81.0% power (80.5% in the sensitivity analysis), exceeding the target power of 80%. Considering this result and research feasibility, the final sample size was determined to be $N = 9$ per group. The complete power analysis implementation, including R scripts, is available in the Zenodo repository [28].

4.4.5 Multiple Comparison Adjustment

For the nine statistical hypothesis tests included in the primary (one Primary Endpoint $\times$ three conditions) and secondary analyses (two Secondary Endpoints $\times$ three conditions), the false discovery rate (FDR) was controlled using the Benjamini–Hochberg method. The 95% confidence intervals were reported for all statistical hypothesis tests, with the second decimal place processed according to banker's rounding (round half to even).

4.5 Transparency and reproducibility measures

The following comprehensive measures were implemented to ensure transparency and reproducibility: Please refer to Section 9 for further details.

- Pre-publication of the Protocol: This evaluation protocol was registered in a public repository, and a digital object identifier (DOI) was obtained before data acquisition.
- Complete Data Publication: Evaluation task definition files, custom instructions (including rubric-equipped scoring AI agents), all chat logs, and evaluation chat logs were published unedited. All research data, statistical

analysis codes, and original Japanese implementations of custom instructions are permanently accessible through the Zenodo repository [28].

- **Statistical Analysis Reproducibility Assurance:** For statistical analysis using R (programming language), the versions of R and the major libraries used are clearly stated in the paper. Additionally, the power analysis implementation, all experimental data, and all custom instructions used in the S-OPE implementation are published. The protocol was pre-published before research implementation, and the entire evaluation process was published unedited to ensure transparency.

All custom instruction implementations, including the S-OPE configuration, were conducted in Japanese. This language choice is necessary given the core concept of the CORAL framework: “structuring and implementing individual thinking processes.” This is because developers’ thinking processes have been formed and developed in Japanese, their native language. Using Japanese when structuring, verbalizing, and implementing those thinking processes in AI is believed to maximize the preservation of the essential structure and semantic nuances of thought. For international readers, English translations of the Test conditions, Control conditions, and scoring AI agents were provided in public repositories. Please refer to Section 9 for further details.

5 RESULTS

5.1 Descriptive statistics

Table 6 lists the descriptive statistics of the final artifact scores for each condition. The Test condition (S-OPE configuration) showed a median of 30.7 points, achieving the highest score compared with the Control (25), ablation-S (27.3), and ablation-H (26.3) conditions.

Table 6: Descriptive Statistics of Final Artifact Scores by Condition: The Test condition (S-OPE) demonstrates the highest median score (30.7), compared with the Control (25.0), Ablation-S (27.3), and Ablation-H (26.3) conditions. SD denotes standard deviation.

Condition	N	Median	Mean	SD	Min	Max
Test (S-OPE)	9	30.7	30.7	1.20	29.0	32.7
Control	9	25.0	24.8	1.57	22.0	27.0
Ablation-S	9	27.3	27.4	1.23	25.0	29.0
Ablation-H	9	26.3	26.3	1.30	25.0	28.3

The intra-rater consistency of the AI scoring agent was high, with a mean SD of 1.74 points (range: 0–4.36) across the 36 reports’ three evaluations per report.

Figure 6 shows the distribution of the final artifact scores for each condition in a box plot. The Test condition showed a clearly higher score distribution compared to all other conditions, visually confirming a minimal overlap in distributions.

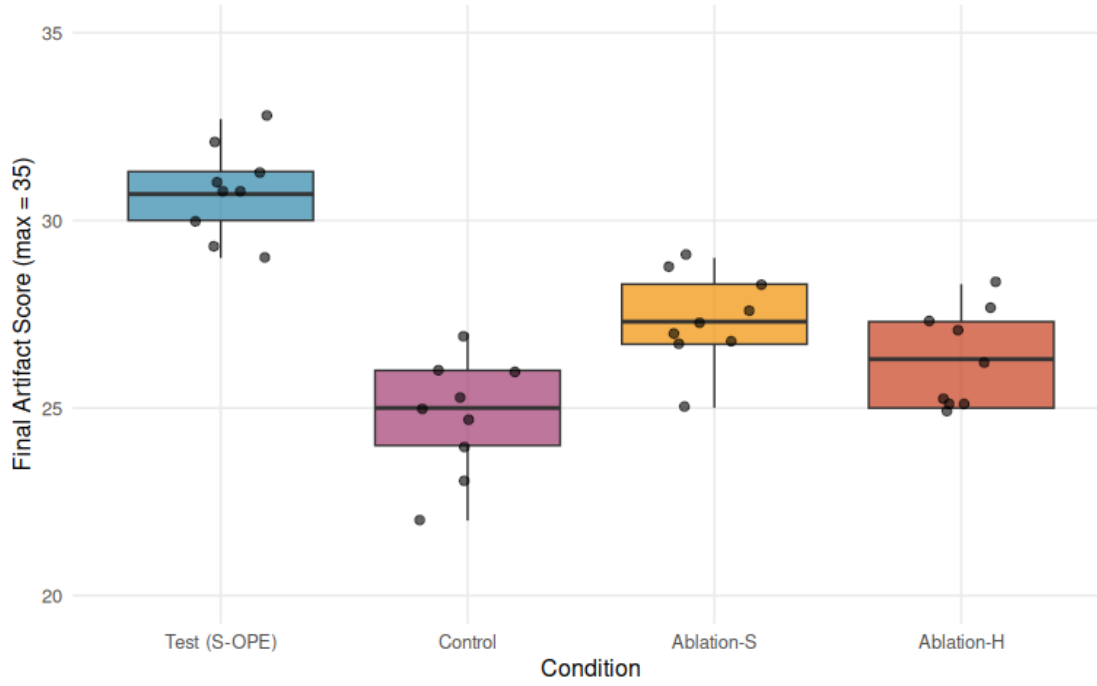


Figure 6: Distribution of Final Artifact Scores by Condition: This box plot shows score distributions across the four experimental conditions. Boxes represent the interquartile range (IQR, Q1–Q3), the horizontal line within each box indicates the median, and whiskers extend to the minimum and maximum values within 1.5xIQR. The Test condition (S-OPE) demonstrates consistently higher scores with minimal overlap relative to the other conditions. Statistical comparisons were performed using two-sided Mann–Whitney U tests ($\alpha = 0.05$), applying Benjamini–Hochberg FDR correction for multiple comparisons.

5.2 Order effects verification

Figure 7 shows the relationship between the execution order and the final artifact scores for each condition. This visualization confirms the absence of systematic order effects (learning or fatigue effects) when a Latin square counterbalancing design was used for control.

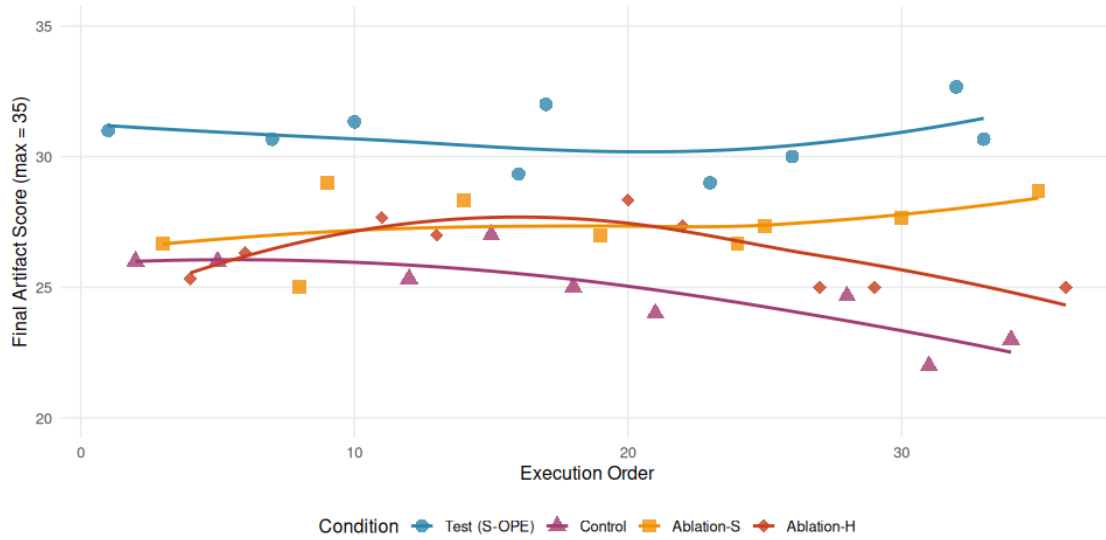


Figure 7: Verification of Order Effects: This scatter plot shows final artifact scores plotted against execution order for each condition. Points are colored by condition and connected with LOESS (Locally Estimated Scatterplot Smoothing) curves to reveal potential systematic trends. The absence of consistent upward (learning effect) or downward (fatigue effect) trends across conditions indicates that the Latin square counterbalancing effectively controlled for order effects.

The scatter plot with LOESS smoothing lines revealed no systematic order effects across conditions. If learning (score improvement over time) or fatigue effects (score decline over time) were present, all conditions would show consistent upward and downward trends. Instead, the observed patterns differed by condition: the Test condition maintained stable scores of approximately 30 points throughout the study period, Ablation-S and Ablation-H showed relatively flat trends, and the Control condition showed a slight downward slope. Importantly, this variation in trends across conditions, rather than a uniform trend, indicates that Latin square counterbalancing successfully controls order effects. The slight downward trend observed in the Control condition was attributable to the inherent variability of that condition rather than to a systematic fatigue effect, as the other conditions did not exhibit similar patterns. Therefore, the condition differences observed in the primary analysis can be attributed to the experimental conditions themselves, rather than to order effects.

5.3 Outstanding evaluation acquisition patterns

Figure 8 shows the pattern of achieving 5-point (Outstanding) evaluations across the seven evaluation dimensions by condition. This visualization reveals which dimensions are more likely to achieve outstanding evaluations and how the patterns differ between conditions.

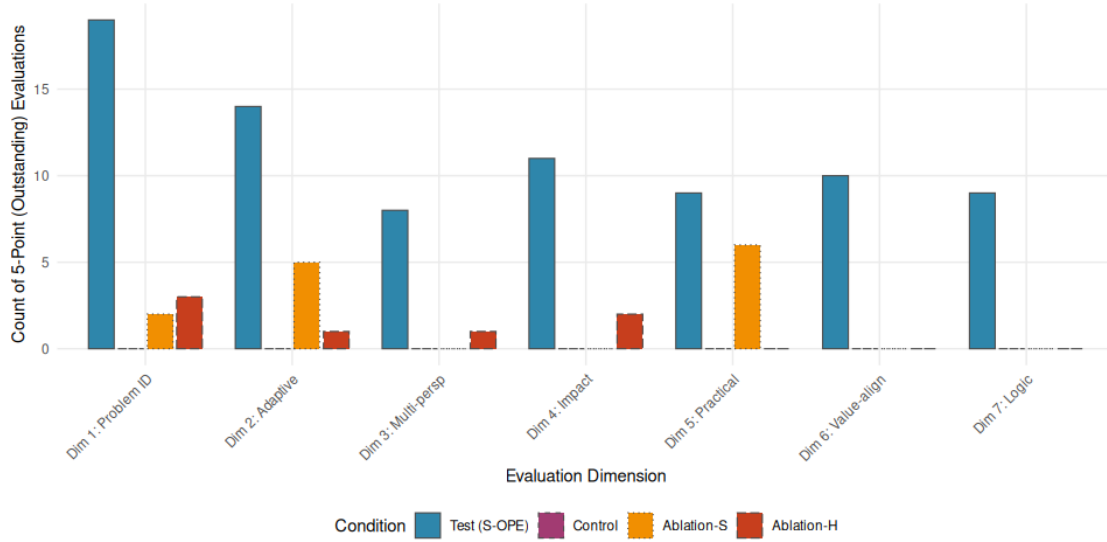


Figure 8: Outstanding (5-Point) Evaluation Acquisition Patterns by Dimension and Condition: This figure depicts the frequency of achieving 5-point evaluations across the seven evaluation dimensions (Dim = Dimension) for each experimental condition. The Test condition (S-OPE) consistently achieves outstanding scores across all dimensions. The Control condition shows almost no 5-point evaluations. The Ablation conditions achieve some 5-point evaluations in specific dimensions: Ablation-S primarily in Dimensions 2 (Adaptive) and 5 (Practical), and Ablation-H primarily in Dimensions 1 (Problem ID) and 4 (Impact).

The visualization reveals that the Test condition achieved 5-point evaluations consistently across all seven dimensions, with particularly high frequencies in dimensions 1 (Essential Problem Identification), 2 (Adaptive Analysis), and 6 (Value-Aligned Output). Notably, although the Control condition achieved almost no 5-point evaluations, the ablation conditions achieved some: Ablation-S (TIDE present, the Sensemaker absent) showed 5-point evaluations primarily in Dimensions 2 and 5, whereas Ablation-H (the Sensemaker present, TIDE absent) showed them primarily in Dimensions 1 and 4. This pattern suggests that each component contributes independently to achieving outstanding evaluations in specific dimensions; however, only the full S-OPE configuration achieves consistent excellence across all dimensions.

5.4 Primary analysis: Mann–Whitney U test

Mann–Whitney U tests (two-sided) were conducted on the final artifact scores between the Test condition and each comparison condition. Results are listed in Table 7.

Table 7: Primary Analysis Results: Mann–Whitney U Test (Two-Sided) Comparing the Test condition (S-OPE) with Each Comparison Condition. All comparisons show statistically significant differences with very large effect sizes ($r > 0.8$ per Cohen’s criteria). The table reports 95% confidence intervals for both the location shift and the common language (CL) effect size (probability of superiority). W denotes the Mann–Whitney U statistic.

Comparison	W	Raw p	95% CI	Effect Size (r)	CL Effect Size
Test vs. Control	81.0	0.0004	[4.3, 7.4]	0.83	1.00
Test vs. Ablation-S	80.5	0.0005	[2.0, 4.6]	0.82	0.99
Test vs. Ablation-H	81.0	0.0004	[3.0, 5.7]	0.83	1.00

In all comparisons, the Test conditions yielded significantly higher scores ($p < 0.001$). Effect sizes (Wilcoxon r) all exceeded 0.8, classified as “very large effects” by Cohen’s criteria.

5.5 Secondary analysis: Excellent and outstanding evaluations

As Secondary Endpoints, the number of 4-point (excellent) and 5-point (outstanding) evaluations achieved in the seven-item evaluation was compared between the conditions. Results are listed in Table 8.

Table 8: Secondary Analysis: Comparison of 4-Point (Excellent) and 5-Point (Outstanding) Evaluation Counts. This table presents the median counts of high-scoring evaluations for each condition (left columns) and the results of Mann–Whitney U tests comparing the Test condition with each comparison condition (right columns). The Test condition exhibits consistently high counts of 5-point evaluations (median = 3.0), whereas the other conditions show a median of 0.0 despite achieving occasional 5-point scores in specific dimensions. The low number of 4-point evaluations for the Test condition reflects elevation to 5-point scores rather than diminished performance.

Condition	Median 4-pt	Median 5-pt	vs Test (4-pt)	vs Test (5-pt)
Test (S-OPE)	3.3	3.0	-	-
Control	4.0	0.0	$p=0.894$	$p<.001^*$
Ablation-S	5.7	0.0	$p=0.011$	$p<.001^*$
Ablation-H	4.7	0.0	$p=0.082$	$p<.001^*$

In achieving 5-point (outstanding) evaluations, the Test condition showed a median of three, demonstrating a notable difference compared to the Control (0), Ablation-S (0), and Ablation-H (0). This result suggests that the S-OPE configuration qualitatively differs in its ability to achieve outstanding evaluations, rather than merely higher average scores.

5.6 Multiple comparison adjustment

FDR adjustment using the Benjamini-Hochberg method was performed on nine tests from the primary (three tests) and secondary analyses (six tests). Results are listed in Table 9.

Table 9: Multiple Comparison Adjustment Using the Benjamini–Hochberg Method: This table presents false discovery rate (FDR)-controlled p-values and 95% confidence intervals for all nine statistical tests (three primary endpoint comparisons and six secondary endpoint comparisons). All primary endpoint comparisons and all 5-point evaluation comparisons remain statistically significant after FDR adjustment (adjusted $p < 0.05$, denoted by *). The 4-point evaluation comparisons are not statistically significant, consistent with the Test condition achieving predominantly 5-point rather than 4-point evaluations.

Test	Raw p	Adjusted p	95% CI	Significant
Primary: Test vs. Control	0.0004	0.0009	[4.3, 7.4]	*
Primary: Test vs. Ablation-S	0.0005	0.0009	[2.0, 4.6]	*
Primary: Test vs. Ablation-H	0.0004	0.0009	[3.0, 5.7]	*
4-point: Test vs. Control	0.8938	0.8938	[-1.7, 1.3]	
4-point: Test vs. Ablation-S	0.0113	0.0145	[-2.7, -0.3]	*
4-point: Test vs. Ablation-H	0.0824	0.0926	[-2.3, 0.3]	
5-point: Test vs. Control	0.0002	0.0009	[2.7, 3.7]	*
5-point: Test vs. Ablation-S	0.0009	0.0014	[1.3, 3.3]	*
5-point: Test vs. Ablation-H	0.0003	0.0009	[1.7, 3.7]	*

Even after FDR adjustment, statistical significance was maintained for all Primary Endpoint comparisons (Test vs. Control, Test vs. Ablation-S, Test vs. Ablation-H) and all 5-point evaluation comparisons (adjusted $p < 0.001$). No significant difference was observed for 4-point evaluations; however, this was attributed to the Test condition achieving 5-point evaluations more frequently than 4-point evaluations.

5.7 Exploratory analysis: Item-specific score distributions

As an exploratory analysis, Table 10 lists the score distributions (median and IQR) for each of the seven evaluation dimensions by condition. This descriptive comparison revealed the dimensions that showed the largest differences between the conditions.

Table 10: Exploratory Analysis: Score Distributions by Evaluation Dimension and Condition. This table presents the median and interquartile range (IQR) for each of the seven evaluation dimensions (0–5 scale) across all four conditions. The Test condition consistently exhibits higher median scores across all dimensions. Note: This analysis is exploratory and intended for hypothesis generation; no statistical tests were performed at the individual dimension level.

Dimension	Test	Control	Ablation-S	Ablation-H
1. Problem Identification	5.0 (4.0–5.0)	3.0 (3.0–4.0)	4.0 (4.0–4.0)	4.0 (4.0–4.0)
2. Adaptive Analysis	5.0 (4.0–5.0)	4.0 (4.0–4.0)	4.0 (4.0–4.0)	4.0 (4.0–4.0)
3. Multi-perspective	4.0 (4.0–5.0)	3.0 (3.0–3.0)	4.0 (3.0–4.0)	3.0 (3.0–4.0)
4. Impact Evaluation	4.0 (4.0–5.0)	4.0 (3.0–4.0)	4.0 (4.0–4.0)	4.0 (3.5–4.0)
5. Practical Solution	4.0 (4.0–5.0)	4.0 (4.0–4.0)	4.0 (4.0–4.0)	4.0 (4.0–4.0)
6. Value Alignment	4.0 (4.0–5.0)	3.0 (3.0–4.0)	4.0 (3.0–4.0)	4.0 (3.0–4.0)
7. Logic and Reliability	4.0 (4.0–5.0)	4.0 (3.0–4.0)	4.0 (4.0–4.0)	4.0 (3.0–4.0)

Figure 9 presents the score distributions as a heatmap showing the frequency of each score (0–5) for each dimension and condition.

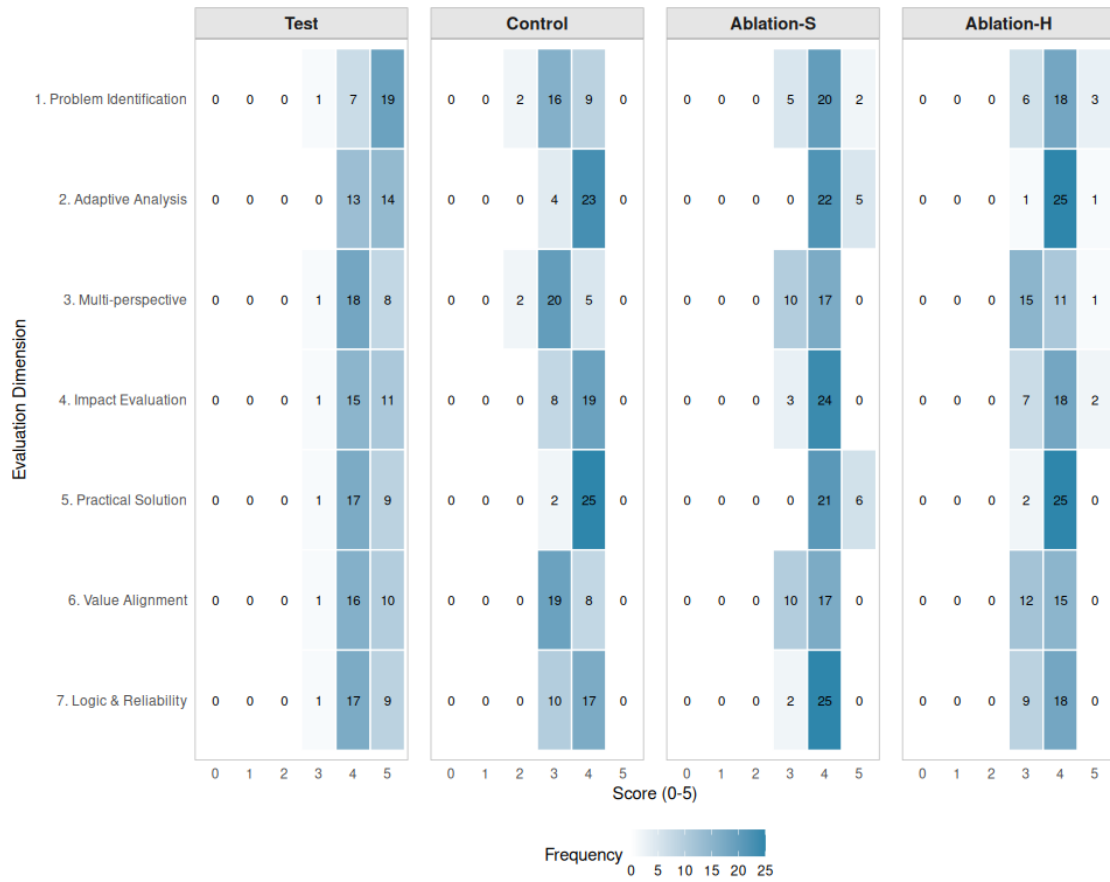


Figure 9: Score Distribution Heatmap by Dimension and Condition: Each Cell shows the frequency of scores (0–5 points) for each evaluation dimension across all dimensions. Darker colors indicate higher frequencies. The Test condition shows a concentration of scores in the 4–5 range across dimensions, whereas the other conditions exhibit greater dispersion, with more scores clustered in the 3–4 range.

The exploratory analysis revealed that the Test condition achieved higher median scores across all seven dimensions than the other conditions. The heatmap indicates that the Test condition has a concentration of scores in the 4–5 point range, whereas the other conditions exhibit more variability, with scores distributed across a range of 3–4 points. Notably, the IQR patterns reveal different stability characteristics: the Test condition shows consistent IQRs of 4.0–5.0 across dimensions, indicating stable high performance, whereas Ablation-S showed zero variance (IQR = 4.0–4.0) in most dimensions (5 of 7) and Ablation-H in some (3 of 7), suggesting a “ceiling effect” at 4 points where they cannot break through to outstanding evaluations. This pattern is consistent with the primary analysis results and suggests that the effectiveness of the S-OPE configuration is not limited to specific evaluation aspects but extends broadly across multiple cognitive dimensions.

5.8 Summary of results

The following key findings were obtained from the quantitative evaluation in this study:

1. **Primary Endpoint:** The S-OPE configuration (Test condition) showed significantly higher scores compared to all comparison conditions in the final artifact scores (FDR-adjusted $p < 0.001$, effect size $r > 0.8$).
2. **Comparison with Control Condition:** The S-OPE configuration demonstrated a median difference of 5.7 points (30.7 vs. 25) compared to the Control condition, integrating existing advanced prompt engineering techniques (CoT + ReAct + Self-reflection).
3. **Ablation Analysis:** Removal of both Sensemaker Cell (strategic dialogue function) and the TIDE protocol (context inheritance function) resulted in significant score decreases. This indicates that both functions of the CORAL framework contribute independently of the effectiveness.
4. **Qualitative Superiority:** The Test condition achieved consistently high 5-point (outstanding) evaluations across all dimensions (median 3), whereas the other conditions showed a median of 0 (Control: 0, ablation-S: 0, ablation-H: 0), despite achieving some 5-point evaluations in specific dimensions. This demonstrates that the S-OPE configuration achieves qualitatively different outcomes and not just quantitative superiority.

6 DISCUSSION

6.1 Quantitative evaluation

The quantitative evaluation results of this study provide evidence that supports the effectiveness of the CORAL framework from multiple perspectives.

The median difference of 5.7 points (30.7 vs. 25, out of 35 points; Table 6) that the Test condition (S-OPE configuration) showed against the Control condition (integrated custom instruction) exceeded the minimum detectable effect size (3 points) set in the power analysis, representing a practically meaningful difference. The effect size $r = 0.83$ is classified as a “very large effect” (Table 7), statistically supporting that the S-OPE configuration’s superiority was not caused by chance. The score distributions visualized in Figure 6 show minimal overlap between conditions, confirming that the observed differences are not attributable to outliers but reflect consistent performance gaps.

The ablation analysis results confirm the validity of the design principles of the CORAL framework. Both the Ablation-S condition (median 27.3), with the Sensemaker Cell removed, and the Ablation-H condition (median 26.3), with the TIDE protocol context inheritance definition removed, showed significantly lower scores than the Test condition. This indicates that the essential inquiry function, facilitated through strategic dialogue (Sensemaker), and the context inheritance function, which maintains semantic coherence between Cells (TIDE), contribute independently to the effectiveness of the CORAL framework.

Integration Effect (Synergy) Analysis: The integration effect of components can be estimated from ablation analysis data. The difference between the ablation-S condition (Sensemaker absent, TIDE present) and the Control condition is +2.3 points (27.3 – 25), which can be interpreted as the standalone effect of the TIDE protocol. Similarly, the difference between the Ablation-H condition (Sensemaker present, TIDE absent) and the Control condition is +1.3 points (26.3 – 25), which can be interpreted as the standalone effect of Sensemaker. The simple sum of these was +3.6 points; the actual difference between the Test and Control condition was +5.7 points. This difference (approximately 2 points) suggests an interaction effect that occurs when the Sensemaker and TIDE protocols are integrated. That is, the technical value of the CORAL framework may exceed the sum of the effects of its individual components. However, this integration effect

estimation is an approximate calculation that assumes independence between ablation conditions, and additional experiments with a 2×2 factorial design are required for a rigorous interaction analysis.

The pattern of achieving 5-point (outstanding) evaluations was noteworthy (Table 8). The Test condition achieved consistently high 5-point evaluations across all dimensions (median 3), whereas all other conditions had a median of zero (Control: 0, ablation-S: 0, ablation-H: 0). Although the Ablation conditions achieved some 5-point evaluations in specific dimensions (Figure 8), only the Test conditions consistently achieved outstanding evaluations. This result indicates that the S-OPE configuration achieved the highest level of evaluation in the evaluation criteria, not just higher average scores. For the 5-point evaluation, the protocol’s meta-criteria require “meeting 4-point criteria plus additionally demonstrating any of strategic thinking, innovativeness, essential connection to mission, or deep insight,” suggesting that the S-OPE configuration has functionality to support these higher-order cognitive processes. The statistical significance of these findings was confirmed even after the FDR adjustment for multiple comparisons (Table 9), thereby strengthening the reliability of the results.

However, for 4-point (Excellent) evaluations, the Test condition actually had fewer (median 3.3333333 vs. Control: 4, Ablation-S: 5.6666667, Ablation-H: 4.6666667). This is the flip side of the Test condition’s tendency to achieve five points rather than four points, consistent with the S-OPE configuration’s design to promote level-up from “Excellent” to “Outstanding.”

Implications for Practitioners: When aiming for high-quality outcomes in complex problem-solving tasks, the following two points are effective: (1) introducing an essential inquiry phase through strategic dialogue and (2) designing semantic context inheritance between processing units. These can serve as design principles applicable to intellectual work that utilizes AI in general, and are not limited to the CORAL framework.

Validity of Order Effects Control: The verification of order effects (Figure 7) confirmed that the Latin square counterbalancing implemented in this study successfully controlled for potential learning and fatigue effects. If systematic order effects exist, then all conditions would exhibit similar upward (learning) or downward (fatigue) trends. The absence of such uniform trends across the conditions supports the internal validity of this study. The slight downward slope observed only in the Control condition likely reflects the inherent inconsistency of that condition, in which performance depends more heavily on momentary prompt interpretation than on experimenter fatigue. This interpretation is consistent with the higher variability observed in the descriptive statistics for the control condition. Conversely, the stability of the Test condition scores across the study period suggests that the structured approach of the S-OPE configuration provides more consistent outcomes regardless of when the task is executed.

6.2 Iterative improvement observed in the development process

The development of the S-OPE configuration (Section 3.3) documents the iterative improvement process of the CORAL framework. In particular, the transition period from the integration experiment failure on April 28, 2025, to the emergence of the TIDE protocol on May 9, 2025, provides a concrete example of double-loop learning.

By analyzing the S-OPE development process from the perspective of Argyris and Schon’s [24] double-loop learning theory, a gradual evolution from single-loop learning to double-loop learning was observed.

Functional improvements in early development (January to April 27, 2025) corresponded mainly to single-loop learning. The introduction of strategic dialogue functionality (April 16) and the addition of orchestration functionality (April 27) were functional extensions within the existing design philosophy, remaining at the improvement of problem-solving “methods.”

However, the attempt on April 28 to integrate multiple Executors into a single file and its failure became a decisive trigger, forcing a review of the system’s “fundamental assumptions.” This attempt was made to reduce complexity by decreasing the number of custom instructions and overall description volume. Behind this was an attempt to practice the tacit knowledge included in existing prompt engineering techniques: “achieve objectives with less description in custom instructions and avoid attention control breakdown.”

This integration experiment failed, as the LLM could no longer grasp the overall instructions, and CORAL as a whole became unstable. This was a case in which integrating multiple Executors created a large single custom instruction as a result, and the descriptions attempting to control it inadvertently increased the descriptions of other custom instructions, ultimately failing to reduce complexity and causing an attention control breakdown. Importantly, this failure led to the discovery of the necessity of Cell division (see Section 2.3.3) and the importance of contextual inheritance.

First, if one intends to enhance thinking by structuring, verbalizing, and implementing one’s thinking processes as AI reasoning processes, the description precision must be increased. This inevitably increases the volume of custom instruction descriptions, making custom instruction management and attention control difficult. Therefore, dividing custom instructions into functional units is necessary, and coordination between the divided Cells is essential. Handling “thinking” interactively with users requires sequential processing rather than MAS-like distributed processing, and processing tasks between multiple Cells while switching abstraction levels or functions should be done under a consistent context. Hence, “context inheritance” that maintains semantic coherence as task handover becomes necessary.

This can be interpreted as double-loop learning, where questioning of fundamental assumptions— “Why was that solution chosen?” clarified the technical challenges inherent in the core concept of structuring and implementing thinking processes leading to solutions. The rapid increase in the description volume from 14,745 to 36,223 characters during the process leading to the emergence of the TIDE protocol on May 9 indicates that the emergence of the TIDE established the Cell concept, solved the technical challenge of attention control breakdown owing to large custom instructions, and contributed to completing the structuring and implementation of thinking processes as a practical functioning CORAL implementation.

The most important insight into this development process is the discovery that structuring and implementing thinking processes is not a pre-designed transformation process but a dynamic process that emerges through human–AI interaction. According to the development records, the establishment of this design policy (May 9) occurred simultaneously with the technical implementation of the TIDE protocol, indicating that theory and implementation developed in parallel. This improvement process can be understood as the cyclical structure of “forethought phase -> performance phase -> self-reflection phase” in Zimmerman’s [2] self-regulated learning theory. The cycles of Cell design (forethought), AI execution (performance), and result observation and modification (self-reflection) were iterated, and the developer refined the Cell descriptions, resulting in improved AI reasoning processes. The records show an iterative development process in which the developer gained new insights from AI behavior to develop Cell descriptions, and the refined Cells subsequently improved the quality of AI reasoning.

6.3 Integrated theoretical implications

By integrating the quantitative evaluation results and development process observation records, the theoretical contributions of the CORAL framework to human–AI collaboration became clearer.

First, the effectiveness of the S-OPE configuration, as demonstrated in the quantitative evaluation, shows that the approach of “structuring and implementing individual thinking processes” functions as a technical implementation that

produces demonstrable effects, not just a theoretical construct. The fact that the Test condition showed statistically significant superiority over all comparison conditions indicates that structuring user cognitive characteristics and implementing them as AI reasoning processes can achieve outcomes exceeding those of general-purpose prompt engineering techniques (Control condition). Importantly, this comparison is not simply about which “prompt” is better. Complex strategic dialogue functions, such as the Sensemaker Cell (approximately 4,500 characters, including multiple thinking patterns and staged dialogue processes), do not function when integrated into a single prompt owing to attention control breakdown (see Section 2.3.3). Therefore, the difference between Test and Control conditions reflects “the value of functions made possible only through Cell division architecture,” not “ingenuity in prompt engineering.”

The ablation analysis results reveal the independent contributions of each element constituting the design principles of the CORAL framework. The performance decreased from removing the Sensemaker Cell, indicating that strategic dialogue functionality contributes to artifact quality. However, whether this effect was caused by the “introduction of desirable difficulties” as intended by the design or other mechanisms was not verified in this study. Additionally, the performance decreased from removing the TIDE protocol’s context inheritance definition, indicating that context inheritance structured across the six perspectives contributes to artifact quality. The dimension-specific patterns of 5-point evaluations in the ablation conditions (Figure 8) provide additional evidence for these independent contributions: Ablation-S achieved some outstanding evaluations in dimensions related to adaptive analysis and practical solutions, whereas Ablation-H showed them in problem identification and impact evaluation dimensions. However, only the full S-OPE configuration achieved consistently outstanding evaluations across all dimensions, suggesting that the integration of both components is necessary for comprehensive excellence. The IQR patterns in Table 10 and the score frequency distributions visualized in Figure 9 further support this interpretation: Ablation-S showed zero variance (IQR = 4.0–4.0) in most dimensions (5 of 7) and Ablation-H in some (3 of 7), indicating a “ceiling effect” at 4 points, whereas the Test condition maintained IQRs of 4.0–5.0 across all dimensions, demonstrating consistent ability to achieve outstanding evaluations.

Considering the double-loop learning case shown in the development process observation record (Section 3.3) and the 5-point evaluation achievement pattern shown in the quantitative evaluation, the mechanism of iterative improvement promoted by the CORAL framework becomes visible. The developer used the failure of the integration experiment (April 28, 2025) as a trigger to fundamentally question the existing assumptions (complexity reduction through custom instruction description volume reduction) and created a new solution called the TIDE protocol. The fact that the S-OPE configuration established as a result of this double-loop learning process became the only condition for achieving the 5-point (Outstanding) evaluation can be interpreted as a concrete outcome of an improvement process based on deep reflection.

These findings suggest a new direction for personalization technologies in the HCI field. While conventional adaptive systems were approaches that adjusted systems “from outside” based on user behavior patterns and preferences, the CORAL framework enables users to structure and verbalize their thinking processes and implement them in AI systems “from within.” The model, which enables continuous improvement based on feedback and potentially promotes user metacognition during this structuring and implementation process, presents a new direction for human–AI collaboration.

Research Agenda for the HCI Community: The findings of this study suggest the following research directions: (1) identification of success factors for structuring and implementing individual thinking processes—what cognitive characteristics, domain knowledge, and expression styles promote effective structuring and implementation; (2) longitudinal measurement of cognitive transformation through structuring and implementation—long-term changes in

metacognitive ability, self-regulated learning ability, and problem-solving skills; and (3) comparison of structuring and implementation patterns across diverse user groups—verification of differences by expert/novice, language/culture, and cognitive style.

6.4 Limitations

This study is constrained by potential bias arising from a single-developer PoC and limited external validity (generalizability). The rationale for the single-developer design and its implications for external validity are detailed in Section 4.1. Although bias mitigation was attempted through the use of dedicated AI agent-based score reports, the developer both designed and implemented the evaluation rubric, making complete elimination of bias difficult. A human rater-based evaluation was not adopted because the developer serving as the rater would introduce an even greater bias, thereby undermining the validity of any comparison.

Constraints owing to language and cultural differences also play important roles. This study examined the CORAL framework’s core concept of structuring and implementing individual thinking processes in Japanese. Considering the close relationship between thinking and language, this implementation may be specialized for the cognitive style of Japanese speakers. If linguistic relativity holds that language structure influences patterns of thought, then the structuring and implementation of thinking processes across different language families (e.g., English, Chinese, Arabic) may exhibit distinct characteristics [29]. Language dependency poses research challenges for verifying the universality of the CORAL framework.

Technical constraints include current LLM context length limitations and response consistency; however, these are expected to improve with future LLM technology development.

Regarding computational costs, notably, the CORAL framework is inherently a high-cost design. The S-OPE configuration uses custom instruction groups of approximately 38,000 characters, and these Cell definitions consume tokens. Furthermore, dialogue-oriented Cells (e.g., the *Sensemaker* Cell) inevitably result in increased token consumption. Therefore, applications in which the CORAL framework is justified are limited to areas where the artifact quality has a value-exceeding cost, such as complex problem-solving and advanced intellectual production. Cost measurement was not included as an evaluation metric in this study, and a high-cost reasoning model was used because this point is self-evident, and the CORAL framework is premised on pursuing artifact quality rather than efficiency. For straightforward tasks or applications in which immediacy is prioritized, single custom instructions such as Control conditions or even simpler prompts are clearly appropriate.

As a constraint of the evaluation methodology in this study, the measurement target was limited to the short-term effects of the final artifact (report) quality. The long-term effects of the iterative improvement cycle, supported by the CORAL framework (improvement in metacognitive ability and qualitative changes in learning processes), were not verified within the scope of this study. The current experimental design measures only artifact quality within a single session and does not capture long-term effects, such as changes in learning curves, continuous improvement in problem-solving ability, and refinement of cognitive styles. This constraint suggests that a longitudinal research design is required to verify long-term effects.

6.5 Evolution of large language model technology and framework validity

Although the evaluation in this study was conducted with a specific LLM, the value of the CORAL framework exhibits structural aspects that are orthogonal to LLM performance improvement. This will be discussed in the following sections.

First, the structural problem of attention control has not been resolved through the evolution of LLM technology. Even if the LLM context length expands, the problem of attention breakdown for large custom instructions remains unsolved. Rather, the “Lost in the Middle” phenomenon, as demonstrated by Liu et al. [26], may worsen with increased context length. Cell division in the CORAL framework (see Section 2.3.3) is an architecture-level solution to this structural problem and is independent of LLM performance.

Second, the difference between “Test and Control” conditions shown in this study’s four-condition comparison reflects the value of the architectural approach of “structuring and implementing individual thinking processes,” rather than LLM performance itself. The result that the removal of both the Sensemaker Cell and TIDE protocol caused a performance decrease in ablation analysis (see Section 6.1) indicates the structural value provided by these components, and this value is not substituted by LLM performance improvement.

Third, the CORAL framework was based on an LLM-independent design philosophy. As stated in Section 2.3.2, the Cells can be implemented on any platform that explicitly invokes custom instructions. Owing to this versatility, the advantages of structuring through Cell division and the TIDE protocol are expected to persist even as more powerful models emerge in the future.

Therefore, despite the rapid evolution of LLM technology, the value of the approach presented by the CORAL framework—“structuring and implementing individual thinking processes”—is considered to have temporal universality.

6.6 Future directions

Future research requires reproducibility verification by other individuals and domain experts, demonstration of utility in various task domains, and comparative research on thinking-process structuring and implementation characteristics in multilingual environments. A particularly important research topic is the verification of the reproducibility of the rapid improvement cycle observed in the S-OPE development. In this study, the development records show cross-domain theory construction and implementation refinement over a short period of several months; however, whether this phenomenon is reproducible in other individuals or different contexts has not been verified.

Through a longitudinal research design, rigorous verification of the long-term changes in continuous human–AI collaboration using the CORAL framework brings user learning curves, metacognitive ability, and refinement of professional thinking processes. This verification was expected to clarify whether the improvement patterns observed in this study are general phenomena beyond individual cases.

Verification of its applicability to diverse cognitive processes is also an important research direction. The S-OPE configuration in this study reflects specific cognitive styles with backgrounds in pharmaceutical research and private company practice. However, elucidation of the structuring and implementation process characteristics in different professional fields (creative activities, technical development, educational practice, etc.) and cognitive characteristics (analytical vs. intuitive, systematic vs. emergent, etc.) is needed.

Technical development directions include the optimization of dynamic relationships between Cells, consideration of parallel processing by multiple LLMs, and implementation of real-time adaptation functions. In particular, the development of technical solutions that deepen personal adaptation while reducing user cognitive load is important for improving the practicality of the CORAL framework.

As a solution direction for the essential high cognitive load required by the CORAL framework, we propose “distributed cognitive load and collective intelligence formation through construction of an ecosystem for reusing

diverse Cells specialized for individuals.” This ecosystem aims to strike a balance between high initial cognitive load and long-term value through a personalized AI collaboration environment.

Theoretical development directions include the formalization of iterative improvement cycles, the development of effect measurement indicators for structuring and implementing cognitive processes, and the construction of integrative frameworks with diverse learning theories (constructivist learning, social learning theory, etc.). Through the development of these theoretical foundations, it is possible to establish a PoC shown by the CORAL framework as a research area with broader academic and practical value.

7 CONCLUSION

This study proposes a CORAL framework as a new approach to structure and verbalize individual thinking processes and implement them as AI reasoning processes, verifying its effectiveness through a PoC with the S-OPE configuration and rigorous quantitative evaluation. Through a controlled comparative study, we demonstrated the feasibility of structuring and implementing individual thinking processes in a natural language. The S-OPE configuration showed significantly higher scores than the Control condition in complex problem-solving tasks (30.7 vs. 25, median difference of 5.7 points, effect size $r = 0.83$, $p < 0.001$), demonstrating effectiveness.

The CORAL framework enables structured human–AI collaboration beyond the conventional “user-tool” relationship. The effectiveness of the CORAL framework in complex problem-solving tasks was empirically demonstrated through statistical comparison using the Mann–Whitney U test and evaluation using a seven-item evaluation rubric.

Future research topics include expanding applications to diverse professional fields, verifying generalizability through large-scale, multi-institutional collaborative research, and longitudinally evaluating the long-term improvement effects. Human–AI collaboration is evolving from the pursuit of efficiency to the amplification of individuality, and the CORAL framework is an implementable solution that concretely demonstrates one direction of this evolution. We hope this study will contribute to the research community toward realizing new forms of intellectual production that maximize individual intellectual assets through continuous improvement cycles.

8 CONFLICT OF INTEREST STATEMENT

The author declares no conflict of interest.

9 DATA AVAILABILITY STATEMENT

9.1 Zenodo DOI reference

All data, analysis codes, and research materials used in this study are available in the Zenodo repository [28].

9.2 English translation of key cells (for international readers)

To facilitate understanding among international readers and the research community, English translations of all custom instructions were published in the aforementioned Zenodo repository (updated version). These translations are provided as reference materials to facilitate understanding of the technical details of the CORAL framework and specific methods for structuring and implementing individual thinking processes in an international research context, as generated through machine translation, and do not guarantee translation accuracy.

9.3 Statistical and execution environment

Statistical analysis was conducted using R version 4.5.1 [30] with the following major libraries: tidyverse 2.0.0 [31], ggplot2 3.5.2 [32], knitr 1.50 [33], and kableExtra 1.4.0 [34].

Regarding the Cursor IDE version, the evaluation protocol was pre-registered, and a power analysis was conducted using Cursor versions 0.50.4 and 0.50.5. The main evaluation trials reported in this study were conducted using Cursor version 1.1.3 to 1.4.3. This version of the progression reflects a normal software update cycle during the research period and does not affect the reproducibility of the experimental design.

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